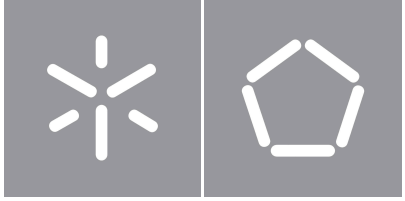


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Computational Discovery of DNA Probe-Graphene Sensors

october 2026



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Computational Discovery of DNA Probe-Graphene Sensors

Master's Dissertation in Bioinformatics

Dissertation supervised by
Dr. João Carneiro
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october 2026

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Acknowledgements

This work received financial support from the PT national funds (FCT/MECI, Fundação para a Ciência e Tecnologia and Ministério da Educação, Ciência e Inovação) through the project UIDB/04423/2020 (CIIMAR), UIDP/04423/2020 (CIIMAR), and UIDB/04050/2020 (UMinho CBMA — https://doi.org/10.54499/UIDB/04050/2020). This work is carried out under contract ref. 2023.15056.TENURE.043 funded by national funds through FCT — Fundação para a Ciência e a Tecnologia, I.P. This work is also funded by national funds through FCT — Fundação para a Ciência e a Tecnologia, within the scope of UID/50016/2025 and LA/P/0076/2020 (https://doi.org/10.54499/LA/P/0076/2020).

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Romeu Alexandre Ribeiro Fernandes

Abstract

Rapid and reliable detection of microbial DNA is central to clinical diagnostics, antimicrobial-resistance surveillance and environmental monitoring, yet the dominant amplification-based methods require laboratory infrastructure that limits their use at the point of care. Graphene field-effect transistor (GFET) biosensors, functionalised with single-stranded DNA recognition elements, offer a compelling label-free alternative, converting the hybridisation of a target sequence at the graphene surface into a measurable electrical signal. The bottleneck of this approach is the recognition sequence: identifying, for each pathogen, probe candidates that are conserved across strains, hybridise favourably, and are free of stable self-structure, without resorting to costly and low-throughput experimental screening of hundreds of candidates.

This dissertation addresses that bottleneck with a reproducible computational pipeline for the *discovery* of DNA probe candidates for GFET biosensors, developed for a panel of clinically relevant bacterial marker genes. From sequences retrieved from the NCBI databases, the pipeline performs automatic length clustering, multiple sequence alignment with MAFFT, and per-column conservation scoring based on the Percentage of Pairwise Identity. Sliding-window candidates are screened with nearest-neighbour thermodynamics (melting temperature, GC content, hairpin and homodimer free energies) and evaluated for secondary structure, condensed into a continuous "No-fold" score. All thresholds are resolved through an organism-aware, layered scheme (gene, species and organism type), so that pathogens with very different genome composition are judged fairly. The highest-quality candidates are prioritised and their structures predicted in three dimensions, and a curated, feature-rich dataset of the best candidates per gene is produced.

Applied to the six-gene panel, with every thermodynamic quantity computed in $1\times$ phosphate-buffered saline, the buffer whose ionic composition is closest to that of physiological fluids, the pipeline reduced 7661 candidate windows to 2439 that passed thermodynamic and structural screening, from which the best probes per gene were predicted in 3D. Alignment-free diversity and rarefaction analyses characterised each target and justified the sampling depth. The result is a prioritised, biophysically sound set of probe

candidates and a reproducible framework that provides the sequence-level foundation for the rational design of DNA probe–graphene GFET biosensors, and that generalises to new genes and organisms with minimal configuration. The pipeline identifies regions conserved within each target gene but does not screen candidates against other organisms; cross-hybridisation with non-target sequences therefore remains to be excluded by a BLAST-based specificity step, and the candidates reported here are candidates for specificity testing rather than a finished panel.

Keywords DNA probe, graphene biosensor, graphene field-effect transistor (GFET), label-free detection, microbial DNA detection, point-of-care diagnostics, probe design, sequence conservation, multiple sequence alignment, melting temperature, secondary structure, No-fold score, 3D structure prediction, Debye length

Resumo

A deteção rápida e fiável de DNA microbiano é central para o diagnóstico clínico, a vigilância da resistência antimicrobiana e a monitorização ambiental, mas os métodos dominantes, baseados em amplificação, exigem infraestrutura laboratorial que limita o seu uso no local de prestação de cuidados. Os biossensores baseados em transístores de efeito de campo de grafeno (GFET), funcionalizados com elementos de reconhecimento de DNA de cadeia simples, oferecem uma alternativa sem marcação apelativa, convertendo a hibridação de uma sequência-alvo à superfície do grafeno num sinal elétrico mensurável. O estrangulamento desta abordagem é a sequência de reconhecimento: identificar, para cada agente patogénico, sondas candidatas que sejam conservadas entre estirpes, hibridem favoravelmente e estejam livres de estrutura secundária estável, sem recorrer à triagem experimental dispendiosa e de baixo débito de centenas de candidatas.

Esta dissertação aborda esse estrangulamento com um pipeline computacional reprodutível para a *descoberta* de sondas de DNA candidatas para biossensores GFET, desenvolvido para um painel de genes marcadores bacterianos clinicamente relevantes. A partir de sequências obtidas nas bases de dados do NCBI, o pipeline realiza a seleção automática do agrupamento de comprimento, o alinhamento múltiplo com o MAFFT e a pontuação de conservação por coluna baseada na Percentagem de Identidade Par-a-Par. As candidatas, geradas por janela deslizante, são filtradas com termodinâmica de vizinhos mais próximos (temperatura de fusão, conteúdo em GC, energias de hairpin e homodímero) e avaliadas quanto à estrutura secundária, condensada num score contínuo "No-fold". Todos os limiares são resolvidos através de um esquema em camadas e sensível ao organismo (gene, espécie e tipo de organismo), para que agentes patogénicos com composição genómica muito distinta sejam avaliados de forma justa. As candidatas de maior qualidade são priorizadas e as suas estruturas previstas em três dimensões, e é produzido um conjunto de dados curado e rico em atributos com as melhores candidatas por gene.

Aplicado ao painel de seis genes, com todas as grandezas termodinâmicas calculadas em tampão fosfato-salino $1\times$, o de composição iónica mais próxima da dos fluidos fisiológicos, o pipeline reduziu 7661 janelas candidatas a 2439 que passaram a triagem termodinâmica e estrutural, a partir das quais

as melhores sondas por gene foram previstas em 3D. Análises de diversidade sem alinhamento e de rarefação caracterizaram cada alvo e justificaram a profundidade de amostragem. O resultado é um conjunto priorizado e biofisicamente sólido de sondas candidatas e uma estrutura reprodutível que constitui o alicerce, ao nível da sequência, para o desenho racional de biossensores GFET de sonda de DNA–grafeno, generalizável a novos genes e organismos com configuração mínima. O pipeline identifica regiões conservadas dentro de cada gene-alvo, mas não compara as candidatas com outros organismos; a hibridação cruzada com sequências não-alvo fica, por isso, por excluir através de uma etapa de especificidade baseada em BLAST, e as candidatas aqui reportadas são candidatas a teste de especificidade e não um painel terminado.

Palavras-chave sonda de DNA, biossensor de grafeno, transístor de efeito de campo de grafeno (GFET), deteção sem marcação, deteção de DNA microbiano, diagnóstico no local de prestação de cuidados, desenho de sondas, conservação de sequência, alinhamento múltiplo de sequências, temperatura de fusão, estrutura secundária, score No-fold, previsão de estrutura 3D, comprimento de Debye

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Acronyms

T_m melting temperature.

AMR Antimicrobial Resistance.

FET Field-Effect Transistor.

FRET Förster Resonance Energy Transfer.

GC Guanine-Cytosine.

GFET Graphene Field-Effect Transistor.

GO Graphene Oxide.

GPU Graphics Processing Unit.

IUPAC International Union of Pure and Applied Chemistry.

MFE Minimum Free Energy.

MSA Multiple Sequence Alignment.

NCBI National Center for Biotechnology Information.

NN Nearest-Neighbour.

PAE Predicted Aligned Error.

PBS Phosphate-Buffered Saline.

PBSE 1-Pyrenebutyric Acid Succinimidyl Ester.

PCR Polymerase Chain Reaction.

pLDDT predicted Local Distance Difference Test.

PoC Point-of-Care.

PPI Percentage of Pairwise Identity.

SELEX Systematic Evolution of Ligands by EXponential enrichment.

ssDNA single-stranded DNA.

WHO World Health Organization.

Chapter 1

Introduction

1.1 Context

Bacterial infections are among the leading causes of death worldwide, and the spread of **Antimicrobial Resistance (AMR)** is steadily eroding the ability to treat them: an estimated 4.95 million deaths were associated with bacterial **AMR** in 2019, of which 1.27 million were directly attributable to it (Murray et al., 2022). Rapid identification of the infecting pathogen, and increasingly of its resistance determinants, is therefore central to guiding therapy, to infection control and to public-health surveillance; several of the species targeted in this work feature on the **WHO** Bacterial Priority Pathogens List (World Health Organization, 2024).

The fast identification of microbial DNA underpins the diagnosis of infection and the tracking of environmental contamination. Culture-based methods are slow, and although the **Polymerase Chain Reaction (PCR)** and related amplification techniques are highly sensitive and specific, they require dedicated equipment, controlled laboratory conditions and lengthy sample preparation, which restricts their use outside the laboratory (Hu et al., 2024). This has driven the development of miniaturised solid-state biosensors, which convert a molecular-recognition event directly into an electrical or electrochemical signal and thereby enable rapid, portable and **Point-of-Care (PoC)** testing (Kozel and Burnham-Marusich, 2017; Krishnan et al., 2023).

Coupling DNA recognition elements to a **Graphene Field-Effect Transistor (GFET)** has been proposed as a particularly promising route to label-free nucleic-acid detection (Green and Norton, 2015). Graphene, a two-dimensional sheet of carbon isolated in 2004 (Novoselov et al., 2004), has its entire conduction channel exposed at its surface and transduces local charge variations with such sensitivity that even a single adsorbed molecule can be detected (Schedin et al., 2007); when it is functionalised with a **single-stranded DNA (ssDNA)** probe, the hybridisation of a complementary microbial target modulates the transistor current in a label-free manner. In practice, the probe is anchored to the graphene

surface through a linker, such as a pyrene derivative that binds non-covalently by π–π stacking, and target hybridisation shifts the transistor's charge-neutrality (Dirac) point; electrolyte-gated graphene devices of this kind have achieved label-free detection of DNA hybridisation down to the attomolar range (Campos et al., 2019). The performance of such devices, and the influence of the sensing interface and ionic conditions on it, is reviewed in detail in Chapter 2.

The specificity and sensitivity of such a device ultimately depend on the immobilised recognition sequence, which must meet several partly competing requirements: it must target a genomic region that is conserved across the clinical strains of the pathogen, so that strain variation does not cause false negatives; it must hybridise strongly and specifically at the assay temperature, which constrains its length, melting temperature and **GC** content; and it must avoid folding into stable hairpins or self-dimers that would compete with hybridisation. These requirements are biophysical and quantifiable, which makes them amenable to systematic computational evaluation.

Building on these experimental developments, a central practical obstacle remains: the high cost and low throughput of experimentally testing hundreds of candidate probes in order to discriminate multiple pathogenic species. This motivates a *complementary computational screening strategy* that can prioritise a small number of promising candidates before any experimental effort is spent.

1.2 Problem statement

The scientific problem addressed in this dissertation is how to computationally *discover and prioritise* DNA probe candidates that recognise specific microbial DNA sequences and are suitable for transduction on a **GFET**. This requires analysing the sequence and structural features — genomic conservation, hybridisation thermodynamics and self-structure — that determine whether a probe will bind its target strongly and specifically across clinical strains, and doing so in a reproducible, automated way that scales across many genes and organisms of very different genome composition (Green and Norton, 2015; Campos et al., 2019).

1.3 Objectives

The main objective is to build a reproducible computational pipeline that turns a target gene into a small set of ranked, biophysically screened **ssDNA** probe candidates for **GFET** biosensing. This decomposes into the following specific objectives:

- select a panel of clinically relevant microbial targets and marker genes, and automate the retrieval and curation of their sequences from the **NCBI** databases;
- identify conserved regions through multiple sequence alignment and per-column conservation scoring, so that the probes detect their target across clinical strains;
- screen candidate probes with well-founded biophysical criteria (melting temperature, GC content, hairpin, homodimer and secondary-structure stability) and combine them into transparent quality scores;
- make the design criteria *organism-aware*, so that targets with very different genome composition (AT-rich or GC-rich) are evaluated fairly;
- prioritise the most promising candidates and characterise them by three-dimensional structure prediction; and
- produce curated, feature-rich datasets of the prioritised probe candidates.

1.4 Target panel

Six marker genes from five clinically important bacterial species (two of them from *Pseudomonas aeruginosa*) were used as the working panel throughout this work (Table 1). Several are established targets for the molecular detection of their species: the thermonuclease gene *nuc* for *Staphylococcus aureus* (Brakstad et al., 1992), the autolysin gene *lytA* for *Streptococcus pneumoniae* (Carvalho et al., 2007), and the outer-membrane lipoprotein gene *oprL* for *Pseudomonas aeruginosa* (De Vos et al., 1997). The panel spans a wide range of genome composition — from AT-rich (*Staphylococcus aureus*, *Haemophilus influenzae*) to GC-rich (*Pseudomonas aeruginosa*) — which makes it a demanding test of a general-purpose design method.

Gene	Organism	GC	Function
nuc	*Staphylococcus aureus*	35.5%	thermonuclease
rmpM	*Neisseria meningitidis*	52.2%	class-4 outer-membrane protein
lytA	*Streptococcus pneumoniae*	44.9%	autolysin
oprL	*Pseudomonas aeruginosa*	63.5%	peptidoglycan-associated lipoprotein
algD	*Pseudomonas aeruginosa*	64.6%	GDP-mannose 6-dehydrogenase
frdB	*Haemophilus influenzae*	39.2%	fumarate reductase Fe–S subunit

Table 1: Target genes and their host organisms used as the working panel. "GC" is the mean GC content of the sequences retrieved for each gene, which spans 29 percentage points across the panel and is what the organism-aware thresholds of Section 3.8 respond to.

1.5 Structure of the dissertation

The remainder of this dissertation is organised as follows. Chapter 2 reviews the state of the art in DNA recognition elements for nucleic-acid biosensing, in graphene **GFET** biosensors, and in the computational methods used to design and evaluate probe sequences. Chapter 3 describes the methods, detailing every stage of the pipeline and the rationale for each design decision and threshold. Chapter 4 presents and discusses the results for the target panel — the screening funnel, the three-dimensional structure prediction, the diversity and rarefaction analyses, and the comparison with a reference probe panel. Chapter 5 draws the main conclusions and outlines future work.

Chapter 2

State of the Art

This chapter reviews the background needed to frame the work. It covers the recognition elements used in nucleic-acid biosensing, graphene **GFET** biosensors for DNA detection, the physics of the DNA–graphene interface, and the computational methods used to design and evaluate probe sequences.

2.1 Recognition elements for nucleic-acid biosensing

Two families of single-stranded DNA recognition element are used in solid-state biosensors, and they differ in how they recognise their target. Aptamers are short oligonucleotides selected *in vitro* by **Systematic Evolution of Ligands by EXponential enrichment (SELEX)** (Tuerk and Gold, 1990; Ellington and Szostak, 1990) that fold into defined three-dimensional structures and bind a target by shape complementarity, with high affinity and specificity (Stephens et al., 2025; Sun et al., 2022). Cell-**SELEX** and related variants extend the approach to whole microbial cells and complex targets, yielding recognition elements for bacterial surface antigens, toxins and viral components (Kolm et al., 2020; Hu et al., 2024). Hybridisation probes, by contrast, recognise a target by Watson–Crick base pairing with a complementary sequence, so their specificity follows directly from the sequence itself rather than from a selection experiment.

The choice between them is dictated by the nature of the target. Aptamers are the appropriate element when no complementary sequence can be written, as for proteins, small molecules or whole cells. When the target is a known stretch of microbial DNA, as it is throughout this work, a complementary hybridisation probe is the natural choice: it can be designed directly from the genome of the pathogen, its affinity is predictable from nearest-neighbour thermodynamics, and it requires no selection campaign. Choosing a hybridisation probe is therefore not a search for a sequence that binds, since any complement will bind, but a search for the right position in the genome. Two questions have to be answered: which gene of the pathogen to target, and which window within that gene to copy. The first determines whether the probe identifies the species at all; the second determines whether it recognises every clinical strain of it and

whether it behaves well as an oligonucleotide. Both are addressed in this dissertation. The two families also impose opposite structural requirements, a contrast that matters for the design criteria of Chapter 3: an aptamer must fold to function, whereas a hybridisation probe must remain unfolded and available to pair with its target. The two also differ in what it costs to obtain them, and the difference does not lie in the synthesis. Once a sequence is known, both are made by the same solid-phase chemistry, which is reproducible and allows terminal modifications to be introduced at will; on that basis aptamers are routinely described as inexpensive by comparison with antibodies (Stephens et al., 2025). The expense of an aptamer lies upstream, in the selection campaign that discovers the sequence: Kolm et al. (2020) required eleven rounds of cell-**SELEX**, with repeated counter-selection against non-target species, to obtain a single aptamer against *Enterococcus faecalis*, and such campaigns are not always successful. A hybridisation probe incurs none of that, because its sequence is read from the genome of the target. This asymmetry, rather than the cost of synthesis, is what makes probes the practical choice when the target is a known nucleic-acid sequence, and it is also what makes them amenable to *in silico* optimisation (Sun et al., 2022).

2.2 Graphene GFET biosensors for DNA detection

Graphene combines very high carrier mobility with a fully surface-exposed conduction channel, so that charges binding at its surface strongly modulate its conductance (Novoselov et al., 2004); the adsorption of even a single molecule can be resolved electrically (Schedin et al., 2007), making it an outstanding transducer for label-free sensing (Krishnan et al., 2023; Green and Norton, 2015), including for nucleic acids at very low concentrations (Hwang et al., 2020). In a **GFET**, the graphene channel is gated electrolytically, and the hybridisation of a target strand to an immobilised probe shifts the transfer characteristic through the added backbone charge. Electrolyte-gated **GFET**s functionalised with DNA probes have reached attomolar, label-free detection of hybridisation (Campos et al., 2019), back-gated architectures have enabled scalable device production (Ping et al., 2016), and microfluidic integration has turned such devices into programmable lab-on-chip DNA sensors (Purwidyantri et al., 2022). Figure 1 shows the arrangement and the quantity that is measured.

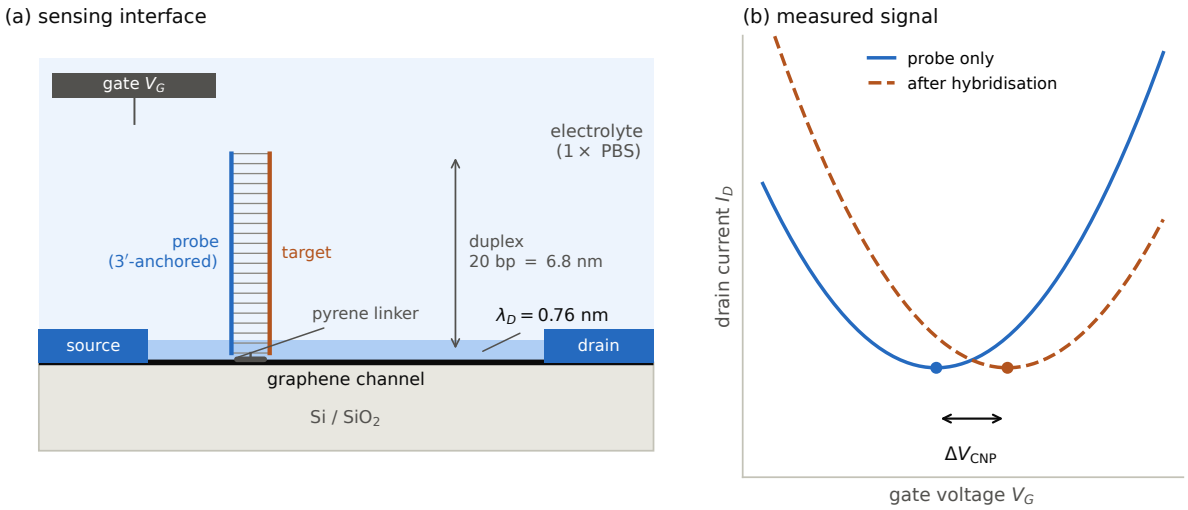


Figure 1: Sensing principle of an electrolyte-gated **GFET**. Left: the probe is held on the graphene channel by a pyrene linker and captures its complementary target, forming a duplex that carries one negative charge per phosphate. The vertical axis is drawn to scale, so the screening length of $1\times$ PBS ($\lambda_D = 0.76$ nm, shaded) can be compared with the 6.8 nm of a 20-base-pair duplex. Right: hybridisation displaces the charge-neutrality point of the transfer characteristic, and that displacement is the signal.

A key operating constraint of any electrolyte-gated **FET** biosensor is the screening of charge by the ions of the solution. Mobile ions in the electrolyte rearrange around any fixed charge and cancel its electric field beyond a characteristic distance, the *Debye length* λ_D, given by

$$\lambda_D = \sqrt{\frac{\varepsilon_r \varepsilon_0 k_B T}{2 N_A e^2 I}},$$

where ε_r and ε_0 are the relative and vacuum permittivities, $k_B T$ the thermal energy, N_A the Avogadro constant, e the elementary charge and I the ionic strength of the solution (Bhattacharyya and Shalev, 2020). Because λ_D scales with $I^{-1/2}$, it shrinks sharply as the buffer becomes more concentrated: only charge within roughly one λ_D of the graphene surface contributes appreciably to the field-effect signal, while charge further away is electrostatically screened. In practice this means that high salt concentration produces strong screening that shortens the effective sensing length and attenuates the signal from charges far from the surface (Purwidyantri et al., 2021); transduction schemes based on target-induced conformational change of the probe have been shown to mitigate this limitation (Nakatsuka et al., 2018). These experimental studies define the device context that the computational design in this dissertation aims to support.

2.3 The DNA–graphene interface

The behaviour of a **GFET** DNA sensor is governed by how DNA interacts with graphene. Single-stranded DNA adsorbs strongly and flat onto graphene through π–π stacking of its exposed nucleobases, whereas double-stranded DNA binds much more weakly because base pairing sequesters the bases (Green and Norton, 2015; Krishnan et al., 2023). This contrast is the basis of many transduction schemes: for example, graphene-oxide **Förster Resonance Energy Transfer (FRET)** sensors, in which a single strand adsorbed on **Graphene Oxide (GO)** is desorbed upon target binding, restoring fluorescence (Lu et al., 2009; Zhu et al., 2015). The contact is close and non-covalent: van der Waals-corrected density-functional calculations place the four nucleobases at 3.16–3.22 Å above the plane, in a regime of strong physisorption (Tayo et al., 2023), consistent with the picture established for the bases on graphene and on hexagonal boron nitride (Lee et al., 2013).

This creates the tension that governs the design of the sensing interface. Charge is only detectable within the screening length, so a strand lying flat on the sheet would place its entire backbone where the transistor can register it; but a strand whose bases are stacked against the graphene has those bases unavailable for Watson–Crick pairing, and therefore cannot capture its target. Adsorption also perturbs folding and kinetics, which matters most for structured recognition elements (Green and Norton, 2015). The practical resolution is to immobilise the probe through a linker rather than let it adsorb: in the devices that define the experimental context of this work the probe carries an amino modification at one terminus and is bound through a pyrene derivative, which itself stacks on the graphene and holds the oligonucleotide off the surface (Campos et al., 2019; Purwidyantri et al., 2021). A useful probe must therefore retain its ability to hybridise while still producing a detectable electronic perturbation at the surface (Green and Norton, 2015), and the geometry that follows from this compromise is measured directly in Section 4.10.

2.4 Computational design and evaluation of probe sequences

Because experimentally screening large numbers of candidate probes is costly and slow, computational methods that link sequence and structure to binding and transduction are increasingly used to design and prioritise candidates (Sun et al., 2022). Several families of criteria are well established. The stability of the probe–target duplex is quantified by its melting temperature (T_m), predicted accurately with the nearest-neighbour thermodynamic model (SantaLucia Jr. and Hicks, 2004), together with GC content and probe length (Wetmur, 1991); tools such as `primer3` implement these calculations (Untergasser

et al., 2012). Self-structure — hairpins, self-dimers and the overall **Minimum Free Energy (MFE)** of the single strand — must be minimised, since a folded probe is unavailable for capture; these energies are obtained from nearest-neighbour folding models (Turner and Mathews, 2010; Zadeh et al., 2011; Timmons, 2020). To detect a pathogen across its strains, conserved genomic regions are identified from a **Multiple Sequence Alignment (MSA)** of many sequences, for which MAFFT is a fast and accurate aligner (Katoh and Standley, 2013); conservation is then scored per column, for example by the **Percentage of Pairwise Identity (PPI)** adopted from the *ViruScope* tool (Lima, 2023).

Beyond sequence, deep-learning methods now predict three-dimensional structure with high accuracy. AlphaFold achieved near-experimental protein predictions (Jumper et al., 2021), and AlphaFold 3 extended this to nucleic-acid-containing complexes with per-residue and pairwise confidence estimates (Abramson et al., 2024); open models such as Boltz-1/2 make these predictions freely runnable on commodity GPUs (Wohlwend et al., 2024). Although calibrated primarily for proteins, the predicted structure and the associated confidence of a short **ssDNA** probe provide a useful, independent check on a computationally selected sequence.

2.5 Summary

Graphene **GFET** devices are an excellent, experimentally proven transducer for label-free DNA detection, but their performance hinges on the recognition sequence. The biophysical criteria for a good probe — conservation, favourable hybridisation thermodynamics and absence of stable self-structure — are individually well understood and supported by mature software, yet they are rarely combined into a single, reproducible and organism-aware pipeline that also ranks the candidates and predicts their structures. The methodology presented in the next chapter is designed to address that gap.

Chapter 3

Methods

The probe-discovery method was implemented as a single, reproducible Python pipeline. Given a target gene, the pipeline proceeds through the stages described below. It is built on Biopython (**NCBI** access and sequence handling), an external MAFFT executable (alignment), `primer3-py` (thermodynamics), `seqfold` (secondary structure) and `pandas` (tabular output); the three-dimensional structure prediction is run separately with Boltz-2 on a **Graphics Processing Unit (GPU)**. Every design threshold is resolved through a layered scheme that can be overridden per gene, per species and per organism type (Section 3.8).

The description follows the order in which the stages run, and falls into three parts. The sequence stages take a gene name and return a ranked list of candidate sequences, by retrieving public records, aligning them, scoring their conservation and screening the resulting windows on biophysical criteria. The best of those candidates are then predicted in three dimensions. Finally, the predicted structures are placed on a graphene sheet and the resulting interface is measured. Three shorter sections close the chapter: the scoring of an existing probe panel by the same metrics, the diversity analysis that justifies the sampling depth, and the handling of intermediate results.

3.1 Sequence retrieval from NCBI

For each target, sequences are retrieved from the **NCBI** nucleotide database through the Entrez interface. An `esearch` query returns up to a configurable number of identifiers (set to 100 in this work; justified in Section 3.13), which are then fetched with `efetch` in FASTA format, in batches of 200 identifiers, with short pauses between requests to respect the **NCBI** rate limit of three requests per second without an API key. Each target defines a primary query combining the gene name, the organism and a sequence-length range; if it returns fewer than two records, the pipeline falls back in turn to one or two broader alternative queries and, failing those, to a fixed RefSeq accession. Records outside a coarse per-gene length range

are discarded at this stage. For a gene with a single usable record, a single-sequence mode is used.

3.2 Length clustering

Public records mix full-length genes with partial fragments and oversized entries, which together produce gap-ridden alignments. An automatic length-clustering step therefore retains only the dominant length cluster: the sequence length whose $\pm 25\%$ band contains the most records is taken as the centre, and only records within that band are kept. The selection is data-driven — no per-gene minimum or maximum length is tuned by hand — so the method transfers to new genes without adjustment. It is applied only when at least four records are available, and a safeguard prevents the set from being reduced below two sequences.

3.3 Multiple sequence alignment

The retained sequences are aligned with MAFFT in its automatic mode (`--auto`). When only one sequence is available the alignment step is bypassed and that sequence is used directly. The alignment of each gene is stored, and the pipeline can restart from the stored alignments instead of querying **NCBI** again. Because the content of the database changes over time, this allows the downstream parameters to be revised without changing the sequences on which the probes are based; all the results reported in this dissertation derive from a single retrieval, processed in this way.

3.4 Conservation scoring (PPI)

With the sequences aligned, the first design criterion can be applied. A probe is useless if the region it targets is present in only part of the strains of the pathogen, since the strains it misses become false negatives; the alignment is what makes that measurable, column by column.

Conservation is scored per alignment column using the **PPI**, adopted from the *ViruScope* tool (Lima, 2023). For a column, the **PPI** is the average identity over all pairs of non-gap bases, computed from the base counts; pairs of ambiguous **IUPAC** bases contribute a partial weight (for example a purine code R scores 0.5 against A or G, and N scores 0.25 against any base). Columns in which fewer than 30% of the sequences have a non-gap base are assigned zero, and only columns with at least two non-gap bases are counted. Because the per-column **PPI** is independent of the window, it is pre-computed once per alignment, which is essential for hundreds of sequences. The conservation of a candidate window

is the mean of the **PPI** of its valid columns. In contrast to primer design, the identity of the terminal $3'$ nucleotides (scored separately as PPI3 in the original tool) is deliberately excluded, because capture probes are not extended by a polymerase and so the fidelity of their $3'$ end is irrelevant.

3.5 Candidate generation

Candidate probes are generated by sliding a window over the alignment. For each start position, taken every 3 columns, windows of every length from 18 to 28 nucleotides are formed. A window is discarded if any of its columns has a gap fraction above the allowed maximum (20% by default), or if its mean conservation is below the target's conservation threshold. The sequence of a surviving window is the per-column consensus (the most frequent non-gap base).

3.6 Thermodynamic screening

Conservation establishes that a region is shared across strains. It says nothing about whether an oligonucleotide copied from that region would hybridise usefully, which depends on the stability of the duplex it forms and on whether the strand instead pairs with itself. Those are thermodynamic questions, and they are answered under the conditions in which the probe is meant to work.

Each candidate is scored with `primer3-py` under the conditions of the reference buffer, $1\times$ **Phosphate-Buffered Saline (PBS)**: 137 mM NaCl, 2.7 mM KCl, 10 mM Na_2HPO_4 and 1.8 mM KH_2PO_4 (Cold Spring Harbor Protocols, 2006), that is, 157 mM Na^+ and 4.5 mM K^+, which `primer3` treats together as a monovalent cation concentration of 161.5 mM. No divalent cations or dNTPs are present, the oligonucleotide concentration is 250 nM and the temperature is $37\,°C$. Four parameters are computed for every candidate: (i) the melting temperature T_m, from the **Nearest-Neighbour (NN)** thermodynamic model (SantaLucia Jr. and Hicks, 2004); (ii) the GC content, the fraction of G and C bases; (iii) the hairpin free energy; and (iv) the homodimer, or self-dimer, free energy. The last two are reported in kcal/mol and only when a structure is found. A candidate passes the basic screen when its T_m and GC content fall within the resolved windows and its hairpin and homodimer free energies are no more negative than their respective limits (Table 2).

The reference conditions are those of the intended application. Among the buffers used for DNA detection with graphene transistors, $1\times$ PBS is the one whose osmolarity and ion concentration approximate those of physiological body fluids (Purwidyantri et al., 2021), and $37\,°C$ is the physiological temperature; for the stability of the probe–target duplex it is also a more demanding condition than room temperature.

Computing every thermodynamic quantity under these conditions gives each threshold a direct meaning. The T_m window is a margin over the hybridisation temperature T: its default, 53–$72\,°\text{C}$, places the T_m between 16 and $35\,°\text{C}$ above $37\,°\text{C}$, and for another assay temperature it can be derived as $[T + 16,\ T + 35]\,°\text{C}$. The lower bound keeps the duplex stable well above the hybridisation temperature; the upper bound excludes probes so stable that hybridisation at that temperature would no longer discriminate imperfectly matched targets (Wetmur, 1991). The hairpin and homodimer limits are absolute free energies of self-structure, so that, evaluated in the reference buffer, they refer to the stability of that structure in the solution in which the probe is used. Because a higher ionic strength stabilises every base-paired structure (SantaLucia Jr. and Hicks, 2004), the same limits are more demanding in $1\times$ PBS than at lower ionic strength, which is the intended consequence of evaluating the probes under their conditions of use.

Parameter	**Default value**	**Basis**
Ionic conditions	$1\times$ PBS	(Purwidyantri et al., 2021; Cold Spring Harbor Protocols, 2006)
Temperature	37 °C	physiological
Probe length	18–28 nt	(Wetmur, 1991)
Melting temperature T_m	53–72 °C	$T + 16$ to $T + 35$ (SantaLucia Jr. and Hicks, 2004)
GC content	40–60%	OligoAnalyzer (Integrated DNA Technologies, 2026)
Hairpin ΔG	≥ -2.0 kcal/mol	OligoAnalyzer (Integrated DNA Technologies, 2026); primer3 (Untergasser et al., 2012)
Homodimer ΔG	≥ -5.0 kcal/mol	OligoAnalyzer (Integrated DNA Technologies, 2026)
Conservation (**PPI**)	≥ 0.85	pan-strain detection (Lima, 2023)
Alignment gap fraction	$\leq 20\%$	alignment quality
Secondary structure ΔG_{MFE}	≥ -2.1 kcal/mol	5th percentile in $1\times$ PBS

Table 2: Global default thresholds, all evaluated in $1\times$ PBS (161.5 mM Na^+ and K^+) at $37\,°\text{C}$. Values are overridden per organism type, species or gene where justified (Section 3.8).

3.7 Secondary structure and the No-fold score

The single-strand secondary structure of each candidate that passed the basic screen is computed with `seqfold`, which returns the **MFE** at $37\,^{\circ}\mathrm{C}$ and the fraction of paired bases. A candidate passes the secondary-structure screen when its **MFE** is no more negative than -2.1 kcal/mol. This value is the fifth percentile of the **MFE** distribution of the candidates that pass the basic screen in the reference buffer (Figure 2). The `seqfold` energies do not take the ionic strength as an input, but the population over which the percentile is taken does depend on it, because the hairpin and homodimer filters that precede this screen are evaluated in $1\times$ PBS; deriving the threshold from that population keeps it consistent with the other criteria. Once derived, the value is fixed, so that the criterion remains reproducible across runs rather than always removing a fixed fraction of the data.

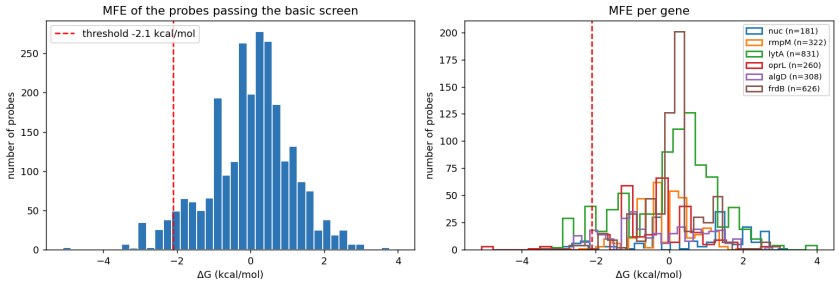


Figure 2: Folding-energy screen: distribution of the `seqfold` secondary-structure free energy (ΔG_{MFE}) of the candidates that pass the basic screen in $1\times$ PBS, overall (left) and per gene (right), with the -2.1 kcal/mol threshold (fifth percentile).

In addition to this pass/fail screen, the three folding energies (hairpin, homodimer and **MFE**) are combined into a continuous *No-fold* score in the range 0–100, following *ViruScope* (Lima, 2023). Each energy is mapped through a logistic penalty,

$$\text{No-fold} = \left(1 - \frac{1}{1+\exp(x)}\right) \times 100, \qquad x = \frac{\Delta G_{\mathrm{cal}} - \Delta G_0/2}{s\,R\,T},$$

with $\Delta G_0 = -7000$ cal/mol, slope $s = 1$, $R = 1.987$ cal mol^{-1}K^{-1} and $T = 310$ K; the per-probe No-fold score is the mean of the scores of the energies that are available. A high No-fold score denotes a probe that is largely free of stable self-structure. This continuous score is used, together with conservation, to rank the candidates rather than merely to filter them.

3.8 Organism-aware parameters

Both screens just described were expressed in terms of the windows into which a candidate's melting temperature and GC content must fall, without saying where those windows come from. This section defines how they are resolved, which is what allows the method to transfer between organisms.

Because the intended application spans pathogens with very different genomes, a single global set of thresholds is inadequate: criteria appropriate for a balanced bacterial genome would unfairly reject an AT-rich or a GC-rich target. All thresholds are therefore resolved from the most specific layer to the most general,

$$\text{gene} \;\rightarrow\; \text{species} \;\rightarrow\; \text{organism type} \;\rightarrow\; \text{global default},$$

where the first layer that defines a parameter wins. Five organism types are distinguished, differing mainly in genome GC content and in variability (Table 3); species-level profiles then fine-tune the values by the GC content of each genome (for example *P. aeruginosa* at ∼67% GC (Stover et al., 2000), *A. fumigatus* at ∼50% GC (Nierman et al., 2005) and the AT-rich *P. falciparum* at ∼19% GC (Gardner et al., 2002)).

Organism type	GC window	Conservation	Max length
Bacteria	40–60%	0.85	28
Virus	35–68%	0.70	30
Fungus	40–60%	0.80	28
Protozoan	15–40%	0.80	30
Host (human)	35–60%	0.85	28

Table 3: Per-type parameter profiles. Viruses use a wider GC window and a more permissive conservation threshold; the protozoan profile is AT-shifted.

When the pipeline encounters an organism for which no profile exists and is run interactively, it prompts the user for the organism type and for each threshold, proposing a suggested value that can be accepted or overridden; the resulting profile is stored and reused in later runs. When run non-interactively, it applies the suggested values without prompting.

Gene profiles. At the most specific layer, each gene of the panel has a profile that departs from the global defaults only where its sequences require it (Table 4). The GC windows follow the measured composition of each target: the AT-rich *nuc* and *frdB* sequences (35.5 and 39.2% GC) accept probes down

to 38% GC, and *nuc* also a T_m down to 52 °C, since A–T pairs contribute less duplex stability than G–C pairs (SantaLucia Jr. and Hicks, 2004); the GC-rich *oprL* and *algD* sequences (63.5 and 64.6% GC, in line with the ∼67% GC of the *P. aeruginosa* genome (Stover et al., 2000)) accept probes up to 70% GC. The relaxed conservation and gap limits of *lytA* follow its allelic diversity (Obregón et al., 2002), which also makes it the most diverse gene of the panel (Table 8), and *algD* is allowed up to 200 records because only 19 are available in **NCBI**. *rmpM*, whose sequences have a balanced composition (52.2% GC) and which is well represented in the database, uses the global defaults. The GC, conservation, gap and length criteria are properties of the sequences and of the alignment and do not depend on the buffer; the T_m and free-energy criteria are evaluated in the reference buffer (Section 3.6).

Gene	Profile	GC	Basis
nuc	$T_m \geq 52$ °C; GC 38–60%	35.5%	AT-rich target
rmpM	global defaults	52.2%	balanced composition
lytA	**PPI** ≥ 0.70; gaps $\leq 40\%$	44.9%	allelic diversity (Obregón et al., 2002)
oprL	GC $\leq 70\%$	63.5%	GC-rich (Stover et al., 2000)
algD	GC $\leq 70\%$; up to 200 records	64.6%	GC-rich; 19 records
frdB	GC 38–60%	39.2%	AT-rich target

Table 4: Gene-level profiles. "GC" is the mean GC content of the retrieved sequences of each gene, gaps excluded. Parameters not listed take the global defaults of Table 2.

3.9 Quality-based selection

Because three-dimensional prediction is comparatively expensive, only the best candidates per gene are carried forward. Rather than ranking by melting temperature — which the thermodynamic screen has already bounded — the surviving candidates are ranked by a *quality* score that averages the two signals most indicative of sequence quality:

$$\text{quality} \;=\; \tfrac{1}{2}\left(\textbf{PPI} + \tfrac{\text{No-fold}}{100}\right).$$

The top-ranked candidates of each gene are exported for 3D structure prediction, and the same ranking is used to export larger curated sets (for example the top 25 per gene) together with their sequence-level features.

3.10 Three-dimensional structure prediction

Every criterion applied so far judges a probe by its sequence. None of them says what that sequence looks like once folded, or how compact the resulting molecule is. Those properties matter at a surface, where only what lies close to it can be sensed, so the next stage predicts the structure itself.

The selected probes are written one per file in the native Boltz-2 input format and submitted to Boltz-2 (Wohlwend et al., 2024), run on a **GPU**, which predicts the three-dimensional structure of each single-stranded sequence together with confidence estimates: an overall confidence, the **predicted Local Distance Difference Test (pLDDT)** and the **Predicted Aligned Error (PAE)** matrix, in the style of AlphaFold 3 (Abramson et al., 2024). As these confidence metrics are calibrated primarily for proteins, the overall confidence and the **pLDDT** are used here as relative, independent indicators of the structural soundness of the short probes. The predictions are merged with the sequence metadata to produce a final ranked shortlist combining sequence quality (conservation and No-fold) with the 3D confidence.

Each selected probe was also predicted in its hybridised state, as a duplex with its full complementary target of the same length (probe as chain A, target as chain B). This is the target format used in the graphene-transistor studies that define the experimental context of this work, in which a 25-nucleotide probe was hybridised to its 25-nucleotide complement (Campos et al., 2019; Purwidyantri et al., 2021).

3.11 Assembly of probe–graphene models

The structures predicted above are of the probe alone, free in solution. In a device the probe is held against a graphene surface, and it is the geometry of that interface, not of the free molecule, which determines how much of the probe's charge the transistor can register. Building that interface requires a model of the surface as well as of the probe.

3.11.1 Generation of the graphene sheet with GOPY

The graphene component was built with **GOPY** (Muraru et al., 2020), a free and open-source Python tool for the automated construction of two-dimensional graphene-based molecular models. Building such a lattice by hand is slow and error-prone, since every atom has to be positioned individually; GOPY generates it programmatically, which makes a sheet of the size needed here practical and reproducible. It writes the model directly in PDB format, so it can be read by standard structural-biology software without conversion. The tool is distributed as a standalone script under the GNU General Public License rather than through a

package index,[1] and is run from the source obtained there, which requires only Python 3 with `numpy` and `scipy`.

For this work only the pristine graphene functionality was used, since the sensing channel modelled here is unfunctionalised graphene. A single layer was generated with the command

```
python GOPY.py generate_PG 120 120 graphene_PG_12nm.pdb
```

where the two numerical arguments are the requested lateral dimensions in Ångström. The resulting sheet was inspected to confirm that it reproduces the expected graphene lattice: it contains 5700 carbon atoms spanning 121.6×119.8 Å, it is strictly planar (all atoms at $z = 0$, with zero dispersion), and the nearest-neighbour carbon–carbon distance is 1.418 Å, the accepted bond length of the graphene honeycomb lattice. Of the 5700 atoms, 5488 are three-coordinated, as expected for the sp^2 interior of the lattice, while the remaining 212 are the two– and one–coordinated atoms that necessarily terminate the edges of a finite, non-periodic sheet. These edge atoms are left unsaturated, as the model is used as a rigid surface for placement rather than in a reactive or dynamical simulation.

The lateral dimensions were not chosen arbitrarily but from the measured footprints of the probes themselves (Section 4.8), so that every probe of the panel fits within the sheet with a margin.

3.11.2 Assembly of the probe–graphene models

Each probe is placed above the sheet by a rigid-body translation, computed with a script built on Biopython (Cock et al., 2009). The probe structure (Boltz-2 CIF) and the sheet (GOPY PDB) are read, and the probe is translated by

$$\Delta = \Big(\bar{x}_g - \bar{x}_p,\ \ \bar{y}_g - \bar{y}_p,\ \ (\bar{z}_g + d) - \min z_p\Big),$$

where $\bar{x}, \bar{y}, \bar{z}$ denote centroid coordinates of the sheet (g) and of the probe (p), and d is the chosen separation. The first two components centre the probe over the sheet in the plane; the third places its lowest atom at a height d above the graphene plane.

The separation d is taken from the literature rather than chosen arbitrarily. Van der Waals-corrected density-functional calculations place physisorbed DNA nucleobases at 3.16–3.22 Å above the graphene plane, a regime of strong physisorption dominated by van der Waals interactions (Tayo et al., 2023), consistent with the physisorption picture established for the four bases on graphene and h-BN (Lee et al., 2013). Accordingly, $d = 3.2$ Å is used as the default, so that the assembled model starts at the reported

[1] https://github.com/Iourarum/GOPY

equilibrium contact distance. The separation is an explicit parameter of the script and can be set to any other value.

The probe's internal conformation and its orientation are left untouched: the operation is a pure translation, so no coordinate is distorted.

The two structures are then merged into a single model and written as one PDB file. Because the PDB format identifies each molecular entity by a single-character chain identifier, and none is reserved for non-biological components such as a graphene layer, an explicit convention was adopted here: the graphene sheet is assigned chain `G` and the probe chain `D`, the initials of *graphene* and *DNA*. The choice is arbitrary in the sense that any identifiers would serve, but keeping the two components in separate, consistently named chains is what allows either of them to be selected unambiguously in downstream tools — for instance to hold the sheet rigid during a docking or minimisation run, or to compute probe–surface distances — and the same convention is used in every model produced.

Before writing, the script verifies that no probe atom falls outside the sheet footprint in x–y and refuses to produce a model with overhanging atoms, so that an undersized sheet cannot pass unnoticed. It also reports the shortest probe–sheet interatomic distance, as a check that the placement introduces no steric clash. The same script builds the entire panel in a single invocation, writing one model per probe together with a table of their measured footprints, heights and closest contacts, so that the assembly stage requires no manual intervention.

It should be stressed that this procedure yields a *starting geometry* — a physically reasonable initial placement of the two components — and not a docked, energy-minimised or dynamically equilibrated structure. It provides the assembled input required by subsequent molecular docking or force-field refinement, but no interaction energy is computed at this stage.

3.11.3 Measurement of the Debye coverage of the models

From each model, the part of the probe that lies close enough to the graphene to escape electrostatic screening is measured. The mean z of the graphene atoms defines the plane of the sheet, and the height of every probe atom above it is computed. Two quantities are reported: the percentage of probe atoms whose height does not exceed the Debye length λ_D, and the number of phosphorus atoms within it. The second is the more direct measure of what the transistor senses, since the charge of DNA is carried by its phosphate groups, one negative charge each.

The values of λ_D were taken from a study of DNA-hybridisation detection with graphene transistors that used these same buffers, where they were calculated with the Debye–Hückel approximation (Purwidyantri

et al., 2021): 0.76 nm for $1\times$ PBS, 2.41 nm for $0.1\times$ PBS and 7.61 nm for $0.01\times$ PBS.

Three geometries are measured. The first is the rigid placement described above, which keeps the orientation predicted by Boltz-2. The second reproduces the immobilisation used in the graphene-transistor studies that define the experimental context of this work, in which the probe carries an amino modification at its $3'$ end and is bound through **1-Pyrenebutyric Acid Succinimidyl Ester (PBSE)**, a linker whose pyrene group adsorbs on graphene by π–π stacking (Campos et al., 2019; Purwidyantri et al., 2021): the probe is tethered by its $3'$ end. To sample the orientations such a tether allows, each predicted structure is given $20\,000$ orientations, drawn as directions distributed uniformly on the sphere; for a flat sheet, the height of every atom depends only on which direction of the molecule points away from the surface, so rotations about that direction need not be sampled. In each orientation the lowest atom is placed at the contact distance of 3.2 Å (Tayo et al., 2023), and the orientation is retained only if that atom belongs to the $3'$-terminal nucleotide, that is, if the probe touches the sheet with its $3'$ end. No linker length is added, so the $3'$ end sits as close to the surface as possible. The median and the maximum over the retained orientations are reported. The third geometry applies the same procedure to the probe–target duplex, anchored by the $3'$ end of the probe, and counts the phosphates of the target strand within λ_D: the charge that hybridisation brings into the sensing volume. As a check on the predicted duplex geometry, its rise per base pair is computed from the midpoints of the C1′ atoms of the paired nucleotides and compared with the 0.34 nm of B-form DNA (Watson and Crick, 1953).

The probes of each gene are finally compared on two independent axes, the Boltz-2 confidence and the median number of phosphates within λ_D in $1\times$ PBS for the anchored probe. The probes on the Pareto front of the two — those for which no other probe of the same gene is at least as good on both axes and better on one — form the final selection. Using the front rather than a weighted sum avoids an arbitrary weighting of two quantities of different nature.

3.12 Integration of a reference probe panel

The pipeline described so far designs probes from scratch. It can also score probes that already exist, which provides a point of comparison external to the method and a test of whether the thresholds behave sensibly on sequences they did not select.

An existing experimental probe panel was integrated into the same workflow for comparison. Each reference probe, supplied as a (`job_name`, `sequence`, `species`) record, is cleaned (uracil converted to thymine, non-ACGT characters removed) and scored with exactly the same metrics as the designed probes,

but by the thresholds of its *own* species and type. Conservation is left empty for these probes, because it is only definable for probes derived from an alignment. The reference probes are stored alongside the designed ones, with a source label, so that the two sets appear side by side with identical columns.

3.13 Diversity and rarefaction analysis

One parameter has so far been stated without justification: the number of sequences requested per gene. Choosing it requires knowing how much variation each gene actually contains, which is also worth knowing in its own right, since it governs how demanding the conservation criterion can be.

The analysis is alignment-free, so that it does not inherit the assumptions of the alignment it is meant to characterise. Each retrieved sequence is reduced to its vector of 4-mer counts over the 256 possible 4-mers, normalised to frequencies. Two distances are then computed for every pair of sequences of a gene: the cosine distance, $1 - \cos\theta$ between the two frequency vectors, which is sensitive to how often each word occurs; and the Jaccard distance, one minus the ratio of the number of 4-mers shared by the two sequences to the number present in either, which ignores frequency and registers only which words occur. The two therefore measure different things, and agreement between them is informative. The fraction of sequences that are distinct is reported alongside, since it detects exact duplicate records.

The sampling depth follows from a rarefaction analysis. For each gene and each N in a fixed schedule, N sequences are drawn at random without replacement, five times independently; the mean pairwise cosine distance and the richness, meaning the number of distinct 4-mers present in the subsample, are averaged over the five draws. Plotting both against N shows the value beyond which further sequences add little, which is the value the retrieval step needs.

3.14 Implementation and reproducibility

All intermediate and final results are written as CSV files. Because **NCBI** does not always return an identical set of records, per-gene counts may vary by a few percent between runs; for strict reproducibility the accession identifiers can be frozen.

The next chapter applies this method to the six-gene panel and reports, stage by stage, what it produced.

Chapter 4

Results and Discussion

This chapter reports the results obtained by applying the pipeline to the six-gene panel, with 100 sequences requested per gene (and a per-gene override of 200 for the data-limited *algD*), with every thermodynamic quantity computed in $1\times$ PBS at $37\,^{\circ}$C (Section 3.6). It follows the flow of the method: sequence retrieval and curation, the screening funnel, three-dimensional structure prediction, the diversity and rarefaction analyses, the effect of the organism-aware parameters, the comparison with the reference panel, the assembly of probe–graphene models, and the curated dataset of prioritised candidates.

4.1 Retrieval and length curation

Table 5 shows, for each gene, the number of sequences returned by **NCBI** and the number retained after length clustering. The clustering step removed partial and oversized records that would otherwise degrade the alignment; its effect is largest for the genes with the most heterogeneous records. Two genes, *oprL* and *algD*, are intrinsically data-limited in **NCBI** (only 24 and 12 sequences survive respectively), which, as shown below, is reflected in their behaviour through the pipeline.

The clustering removed between 0 and 37% of the records depending on the gene. It discarded nothing at all for *frdB*, whose 82 records all fall within the dominant length band, and most for *algD*, which lost seven of its nineteen. The alignment widths that result differ by a factor of five, from the 489 columns of *frdB* to the 2594 of *lytA*: an alignment is as wide as the longest record plus every insertion, so a gene whose records are uniform in length yields a compact alignment, while one with divergent alleles yields a sparse and much wider one. These two extremes recur throughout the chapter, and Section 4.4 shows that they are a property of the genes rather than of the retrieval.

Gene	Retrieved	After clustering	Alignment (cols)
nuc	99	77	517
rmpM	100	76	1311
lytA	95	87	2594
oprL	27	24	770
algD	19	12	1020
frdB	82	82	489

Table 5: Sequences retrieved and retained per gene, and the resulting alignment width.

4.2 Screening funnel

Across the six genes, the pipeline generated 7661 candidate windows and filtered them down to 2439 that passed both the thermodynamic screen and the secondary-structure screen (Table 6). The GC, melting-temperature and homodimer filters remove the largest numbers of windows (1687, 1609 and 1403, respectively), as expected, since they encode the core hybridisation requirements. The homodimer filter is most severe for *oprL*, *rmpM* and *algD*, removing 55, 47 and 41% of the windows that reach it, and mildest for the AT-rich *nuc* (8%). In the GC-rich *P. aeruginosa* genes, the upper bound of the T_m window also removes many windows (583 of the 1360 of *oprL* and 368 of the 1012 of *algD*), since G–C pairs raise the stability of the duplex (SantaLucia Jr. and Hicks, 2004); this is the purpose of that bound, which excludes over-stable probes, and it still leaves 254 and 288 candidates for the two genes. The secondary-structure screen removes only 136 further windows, confirming that most structurally problematic candidates are already eliminated by the thermodynamic filters. Figure 3 places these counts against the stages of the method, so that the procedure of Chapter 3 and its effect on the candidate pool can be read together.

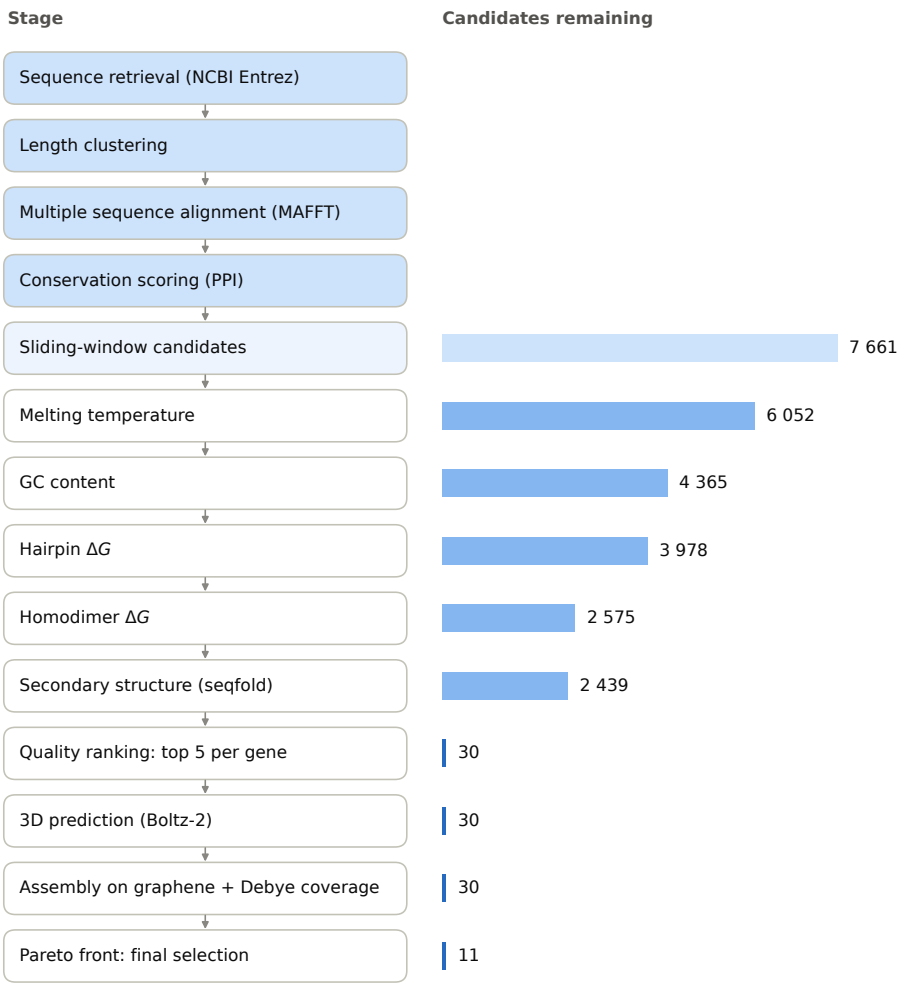


Figure 3: Screening funnel: stages of the pipeline and the number of candidates surviving each one. The first four stages act on sequences rather than on candidate windows, so no count is given for them; the bars begin with the windows produced by the sliding window and end with the final selection. The counts are those of Table 6, obtained in $1\times$ PBS.

Gene	Windows	T_m	GC	Hairpin	Basic	Seqfold
nuc	667	572	212	197	181	164
rmpM	1073	893	683	607	322	320
lytA	1833	1671	1234	1153	876	799
oprL	1360	773	669	576	261	254
algD	1012	644	580	522	308	288
frdB	1716	1499	987	923	627	614
Total	**7661**	**6052**	**4365**	**3978**	**2575**	**2439**

Table 6: Screening funnel in $1\times$ PBS: candidate windows surviving each successive filter. "Basic" is the number passing all four thermodynamic criteria; "Seqfold" additionally passes the secondary-structure screen.

4.3 Three-dimensional structure prediction

The funnel leaves far more candidates than can reasonably be predicted in three dimensions, which is what the quality ranking is for: it decides which sequences receive it.

From the candidates that passed screening, the five highest-quality per gene (30 probes in total) were selected by the quality score and submitted to Boltz-2, each both as a single strand and as a duplex with its complementary target. All 60 predictions completed successfully.

Single strands. By the overall confidence, 4 probes were classified as GOOD (≥ 0.70), 13 as MODERATE (0.60–0.70) and 13 as LOW (below 0.60), with confidences between 0.487 and 0.752 and a mean of 0.610. The four GOOD probes belong to four different genes (Table 7), so the panel does not concentrate its best-predicted structures in a single target. These values are modest in absolute terms, which is expected: the confidence of Boltz-2, like that of AlphaFold 3 on which its metrics are modelled (Abramson et al., 2024; Wohlwend et al., 2024), is calibrated primarily on proteins, and a short unpaired **ssDNA** strand has no single well-defined fold to predict. They are therefore used here as relative indicators among probes of comparable length, not as absolute measures of structural correctness.

Probe	Gene	PPI	No-fold	Confidence	pLDDT
p598	*nuc*	0.995	96.1	0.752	0.896
p5178	*algD*	0.929	91.7	0.738	0.909
p6922	*frdB*	0.993	99.2	0.727	0.828
p2571	*lytA*	0.879	97.2	0.725	0.866

Table 7: The four probes classified as GOOD by the Boltz-2 confidence, with their sequence-level conservation and No-fold score.

Duplexes. Predicted with their complementary targets, the same probes reach substantially higher confidences, from 0.845 to 0.949 (**pLDDT** 0.855–0.985), and all 30 fall in the highest class. The contrast with the single strands follows from what is being predicted: a fully complementary duplex has a defined, canonical structure, whereas an unpaired strand does not. The predicted duplexes are indeed canonical, with a rise of 3.20–3.43 Å per base pair (median 3.28 Å), in agreement with the 3.4 Å of B-form DNA (Watson and Crick, 1953). This is a useful internal check: the structures used in the interface analysis of Section 4.8 reproduce the geometry of real DNA rather than an artefact of the prediction.

Relation to the sequence-level ranking. The 3D confidence is statistically independent of the sequence-level quality that was used to select these probes: across the 30 probes, the Pearson correlation between the confidence and the quality score is $r = -0.01$, and the confidence is likewise unrelated to the conservation ($r = 0.05$) and to the No-fold score ($r = -0.10$). The structure prediction therefore does not restate the sequence ranking: it contributes information of its own. That is why it enters the final selection as a separate axis (Section 4.10.3) rather than being folded into the quality score, where it would have been diluted by two signals with which it does not correlate.

Reproducibility of the predictions. Because the change of reference buffer altered part of the selection (Section 4.2), 9 probe sequences were predicted in both the preliminary and the final run, which allows the run-to-run variability of Boltz-2 to be measured directly. For those sequences the confidence changed by -0.003 on average, and by at most 0.052; the predictions are therefore stable, and differences of this order between probes should not be interpreted. Over the whole panel, the final selection has a slightly higher mean confidence than the preliminary one (0.610 against 0.603) and the same number of GOOD probes, which indicates that screening in the reference buffer did not degrade the structural quality of the candidates it selects.

4.4 Genetic diversity of the targets

The genes did not behave uniformly through retrieval and screening, and part of that is a property of the targets themselves rather than of the thresholds applied to them: how variable each gene is across the strains deposited in **NCBI**. The alignment-free diversity analysis (Table 8) measures that directly. *lytA* is by far the most diverse gene (mean cosine distance 0.187 and 93% unique sequences), consistent with its known allelic diversity in *S. pneumoniae* (Obregón et al., 2002); this justifies the relaxed conservation threshold applied to it. At the opposite extreme, *frdB* is the most conserved and most redundant gene (mean cosine distance 0.025 and only 35% unique sequences, i.e. many exact duplicates in **NCBI**), which is why it yields probes of very high conservation. *algD* has 100% unique sequences but very few of them, which accounts for its instability at low sampling and motivates the automatic length-clustering and the per-gene allowance of more sequences. The cosine and Jaccard distances are complementary — the former weighs 4-mer frequencies, the latter only their presence or absence — and they concur in identifying *frdB* as the most conserved and redundant gene, giving an alignment-independent reading of diversity. Figure 4 shows the same comparison pair by pair: the near-uniform dark field of *frdB* against the mottled one of *lytA*.

These values describe the sequences deposited in **NCBI**, not the strains circulating clinically. Public records accumulate from sequencing efforts that are themselves uneven: outbreak isolates and reference strains are over-represented, and some entries duplicate one another, as the fraction of unique sequences shows directly for *frdB*. The diversity measured here is therefore a lower bound on the diversity a probe would meet in practice, and the conservation thresholds should be read in that light.

Gene	n	Cosine (mean)	Cosine (max)	Jaccard (mean)	Unique (%)
nuc	77	0.059	0.425	0.163	61.0
rmpM	76	0.079	0.247	0.069	50.0
lytA	87	**0.187**	0.440	0.125	93.1
oprL	24	0.050	0.358	0.096	91.7
algD	12	0.097	0.332	0.117	**100.0**
frdB	82	**0.025**	0.052	0.061	**35.4**

Table 8: Alignment-free 4-mer diversity per gene. Bold marks the most diverse (*lytA*) and the most conserved/redundant (*frdB*) genes.

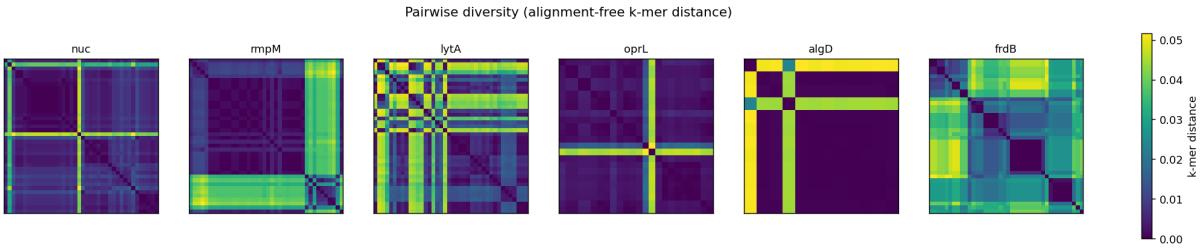


Figure 4: Sequence diversity per gene: pairwise 4-mer distance heatmaps (alignment-free). Each cell is the 4-mer distance between two retrieved sequences of the gene.

4.5 Choice of sampling depth

The rarefaction analysis (Figure 5) showed that both the 4-mer diversity and the 4-mer richness saturate at around $N \approx 50$ sequences for the most demanding gene (*nuc*), and much earlier for the others. Requesting up to 100 sequences per gene therefore captures essentially all of the available diversity with a comfortable margin, while additional sequences add little new information. The true limiting factor is not this cap but the small number of records available in **NCBI** for the data-limited genes.

The distinction matters for how the conservation values of each gene should be read. For *nuc*, *rmpM*, *lytA* and *frdB*, between 76 and 87 sequences survive curation, comfortably past the saturation point, so their conservation scores rest on a sample that further records would not materially change. For *oprL* (24) and *algD* (12) they do not reach it, and the per-gene allowance of 200 records for *algD* changes nothing, because the database holds only 19. The conservation measured for those two genes is therefore the less firmly established of the panel, and this is a limitation of the public record rather than of the sampling strategy.

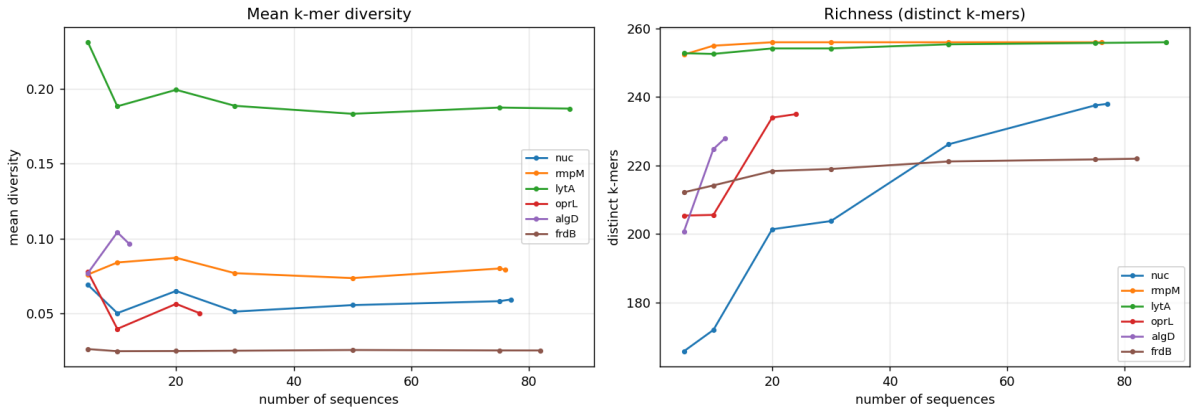


Figure 5: Choice of sampling depth: rarefaction of the mean 4-mer diversity (left) and of the richness, i.e. the number of distinct 4-mers (right), per gene, as a function of the number of subsampled sequences.

4.6 Effect of the organism-aware parameters

The value of resolving thresholds per organism type and species is clearest when scoring the reference panel, which spans bacteria, viruses, a fungus, a protozoan and a human control. When the 74 reference probes were scored with a single global (bacterial) criterion, 33 passed the basic screen; when instead each was scored by the criteria of its own species and type, 42 passed. That net gain of nine is the result of two opposite movements, shown in Figure 6: ten probes are recovered and one is rejected. The ten recovered are those whose genome composition departs from the balanced bacterial case: six from the GC-rich *P. aeruginosa*, two from *K. pneumoniae*, one from the AT-rich *S. aureus* and one from *P. falciparum*. The global GC window rejects the last of these although it is unremarkable for a genome of 19% GC (Gardner et al., 2002). The probe that is lost is the *A. fumigatus* one, whose species profile (42–58% GC) is narrower than the global window it replaces.

The rejection is as informative as the recoveries, because it shows that the layered scheme is not a relaxation applied to awkward targets, but the replacement of a generic criterion by one matched to the composition of each genome, which can tighten the requirement as readily as loosen it.

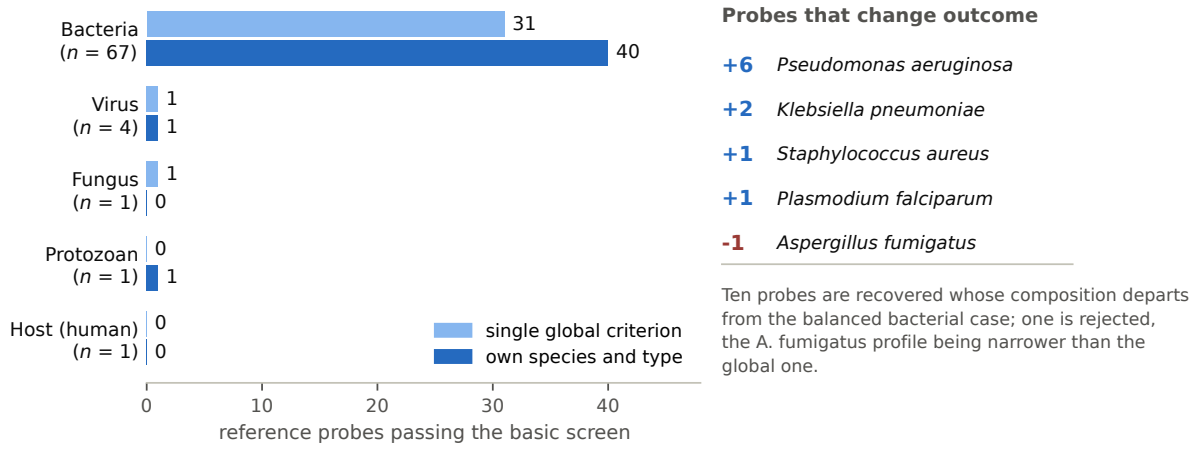


Figure 6: Organism-aware criteria on the reference panel: the 74 reference probes under a single global criterion and under the criteria of each probe's own species and type, by organism type (left), and the probes whose outcome changes (right).

The gene-level profiles have the same effect on the designed probes: 10 of the 30 probes finally selected pass the basic screen only because of the profile of their gene — four of the five *nuc* probes and three each of the *algD* and *frdB* probes — and would have been rejected by the global GC or T_m window although they suit the composition of their target.

4.7 Comparison with the reference panel

Scored with identical metrics, the designed probes and the reference panel can be compared directly. Among the probes that pass the basic screen, the designed probes have a slightly higher mean melting temperature (62.6 versus $61.4\,^{\circ}$C) and a very similar mean GC content (0.49 versus 0.51), while the reference probes have a slightly higher mean No-fold score (84.7 versus 81.4).

None of these differences is large, and that is the useful outcome. The reference probes were selected experimentally, by criteria external to this pipeline, so the fact that the designed probes fall in the same region of melting temperature and composition indicates that the thresholds are not drifting away from what works in practice. The No-fold comparison is the more pointed of the three: the designed probes are screened and ranked partly on that very score, so one would expect them to lead on it, and they do not. Experimentally selected probes are at least as free of self-structure as the ones this method produces, which means the score is measuring a property real probes possess rather than rewarding the output of the method that uses it. Conservation cannot enter the comparison at all, since it is defined only for a

probe derived from an alignment, and the reference probes are not.

4.8 Probe–graphene models

The results reported so far concern the probes as sequences and as isolated structures. None of those criteria describes how a probe sits against the graphene, or how much of it lies where the transistor can sense it, which is what the interface models were built to measure.

All 30 probes predicted in 3D were combined with a graphene sheet to obtain atomistic models of the sensing interface (Section 3.11). The predicted probe structures contain between 357 and 529 atoms (median 403), consistent with single strands of 18–26 nucleotides.

Sizing the sheet. The sheet has to hold every probe without overhang, and its lateral dimensions were therefore set from the probes themselves. The largest in-plane extent of each predicted structure has a median of 44.6 Å and reaches 67.1 Å (Table 9), the probes being predicted in a range of extended and compact conformations. A 4×4 nm sheet, a size that would suffice for a nucleotide-scale model, would have been too small for 19 of the 30 probes, whose atoms would protrude beyond the edges — an artefact that the fit check built into the assembly script rejects rather than writing a model that is geometrically wrong. The 121.6×119.8 Å sheet (5700 carbon atoms) accommodates all 30, the most extended still leaving a margin of 27.2 Å on each side.

Assembled models. The complete set of 30 probe–graphene models was built in a single automated run, each model comprising 6057–6229 atoms; Table 9 summarises their geometry by gene. No model contains a steric clash: across all 30 systems the closest probe–sheet atom pair lies between 3.21 and 3.42 Å, that is, within the physisorption regime reported for nucleobases on graphene (Tayo et al., 2023) and never below it. The small spread above the nominal 3.2 Å separation arises because the probe atom closest to the surface is generally offset laterally from the graphene atoms beneath it, so the true interatomic distance slightly exceeds the vertical one. The probes stand well above the surface, their vertical extent ranging from 23.3 to 59.2 Å; probes of similar length therefore present very different amounts of their structure close to the sensing surface, which is the property quantified in Section 4.10.

Gene	Models	Median extent (Å)	Max. extent (Å)	Height (Å)	Closest contact (Å)
nuc	5	46.1	51.9	25.3–59.2	3.24–3.36
rmpM	5	45.3	46.1	27.5–45.9	3.22–3.26
lytA	5	51.4	67.1	27.2–51.3	3.21–3.42
oprL	5	44.0	58.6	23.8–54.7	3.21–3.32
algD	5	42.3	61.0	23.3–49.0	3.21–3.36
frdB	5	33.1	46.8	34.5–37.6	3.23–3.28
All	**30**	**44.6**	**67.1**	**23.3–59.2**	**3.21–3.42**

Table 9: Geometry of the 30 assembled probe–graphene models. "Extent" is the largest in-plane dimension of the probe, which determined the sheet size; "height" is its vertical extent; and "closest contact" the shortest probe–sheet interatomic distance.

Figure 7 shows the assembled models of the GOOD probes of *algD* and *frdB*, drawn directly from the atomic coordinates of the output files. The side view makes the assembly geometry explicit — the planar graphene lattice, the probe resting above it and the 3.2 Å separation — while the top view shows the probe footprint contained well within the sheet. The two panels also illustrate the conformational contrast noted above: *p5178* is compact, with a vertical extent of 23.3 Å, whereas *p6922* reaches 37.6 Å.

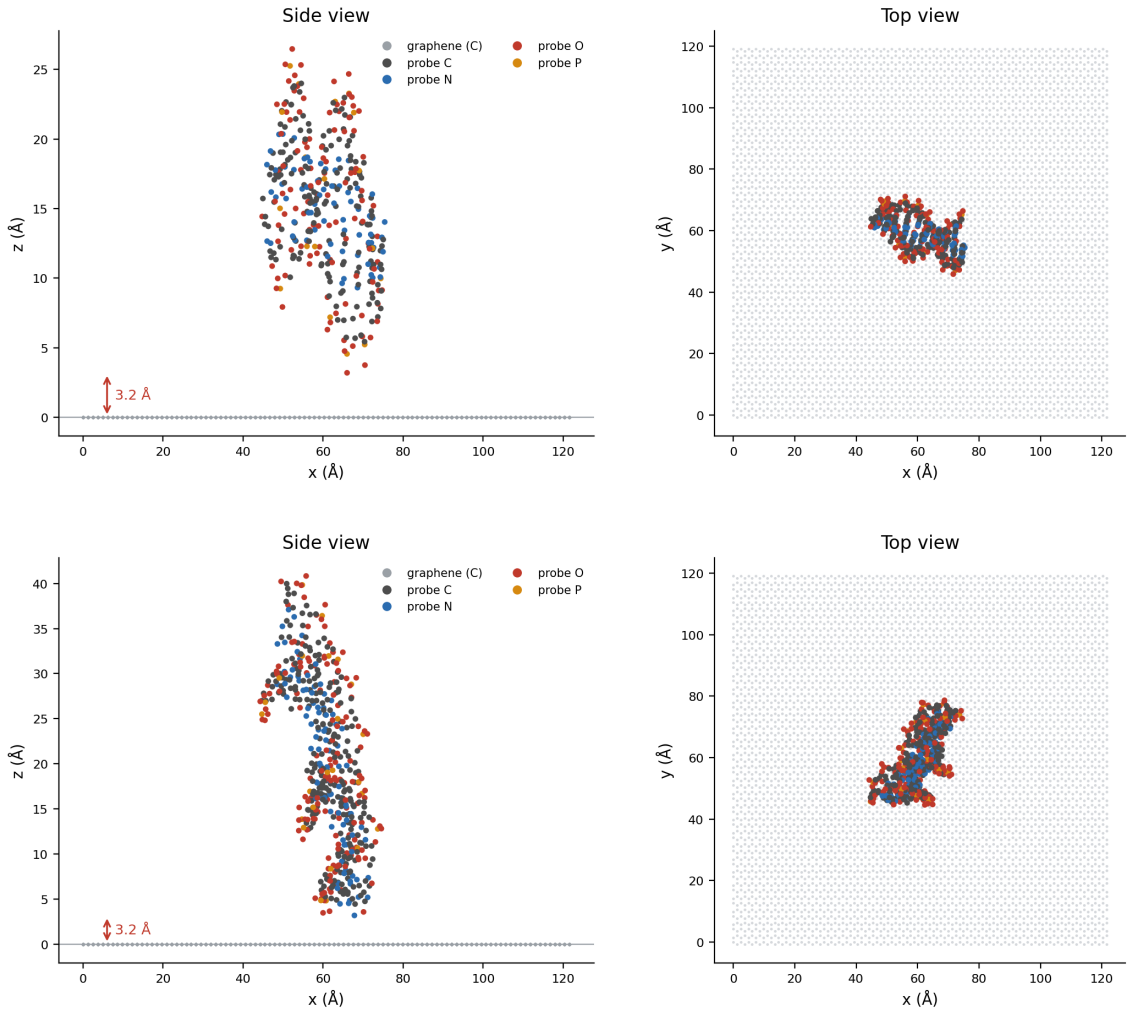


Figure 7: Assembled probe–graphene models for *p5178* (*algD*, top) and *p6922* (*frdB*, bottom), rendered from the model coordinates. Left: side view, with the graphene plane, the probe above it and the 3.2 Å separation indicated. Right: top view, showing the probe footprint within the 121.6×119.8 Å sheet. Probe atoms are coloured by element; graphene carbons are grey.

These assembled structures are starting geometries for subsequent docking or force-field refinement. No interaction energy is computed at this stage, so they support no claim about binding affinity or preferred orientation; what they establish is that the whole panel can be assembled into clash-free, physically sensible interface models by an automated procedure, and that the resulting models are suitable inputs for the refinement stages that follow.

4.9 Why $1\times$ PBS is kept as the reference

A probe designed for $1\times$ PBS will lie largely outside the screening length of that buffer, as Section 4.10 measures. That is not an argument for diluting it, and the reason is dimensional. Figure 8 sets the relevant lengths side by side. In B-form DNA the base pairs are 0.34 nm apart along the helix axis (Watson and Crick, 1953), so the 18- to 26-nucleotide probes selected in this work form duplexes 6.1 to 8.8 nm long (Section 4.10 measures the consequence directly), whereas λ_D is 0.76 nm in $1\times$ PBS: even with the duplex standing on the surface, only about two base pairs lie inside it, and no probe long enough to hybridise specifically would fit. The mismatch is general for electronic biosensors, in which the Debye length under physiological conditions is below 1 nm while a 30-base aptamer extends up to about 10 nm (Kesler et al., 2020). Lying mostly outside λ_D in $1\times$ PBS is therefore a property of the transduction rather than of these particular probes, and shortening them would reduce the stability and specificity of the duplex without bringing it inside.

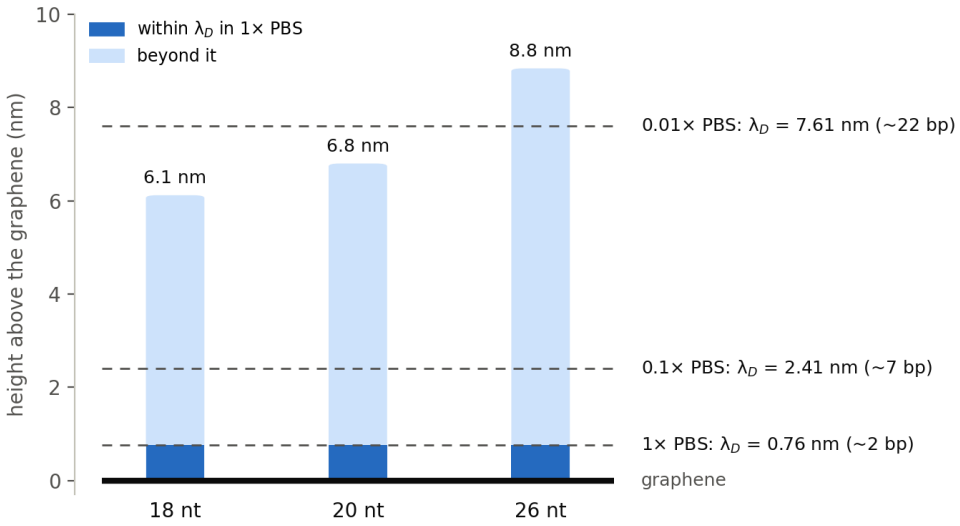


Figure 8: Probe length against the Debye length: a hybridised probe (B-form duplex, 0.34 nm per base pair (Watson and Crick, 1953)) for the shortest, median and longest probes selected (18, 20 and 26 nucleotides), compared on the same vertical scale with the Debye length of the three PBS dilutions (Purwidyantri et al., 2021). The dark segment is the part within λ_D in $1\times$ PBS. The geometry drawn is the most favourable one the duplex could adopt, so the dark segment is an upper bound: the duplex stands perpendicular to the surface and touches it, and no linker is interposed. Any departure from that arrangement lifts the duplex further away. Tilting it raises every base pair above the first, their height becoming the rise along the helix multiplied by the cosine of the tilt; the pyrene linker used to immobilise the probe adds its own length between the surface and the first nucleotide; and a strand adsorbed flat rather than standing, as single-stranded DNA tends to be on graphene (Green and Norton, 2015), would lie within λ_D but would no longer be free to hybridise.

Several further considerations support keeping $1\times$ PBS as the reference condition. It is the medium the device is meant to face: its osmolarity and ion concentration approximate those of physiological body fluids (Purwidyantri et al., 2021), so optimising the probes for a diluted buffer would optimise them for a condition that real samples do not have. Hybridisation is, moreover, still detected in $1\times$ PBS. In the study from which the Debye lengths are taken, hybridisation of a 25-nucleotide target was detected at attomolar concentrations in all three dilutions (Table 10), and a tenfold increase of λ_D, from $1\times$ to $0.01\times$ PBS, only doubled the sensitivity. The Debye length derives from a simplified model of the double layer, valid only for small potentials (Kesler et al., 2020), and the authors of that study attribute the response in $1\times$ PBS to the contribution of the buffer ions to the ionic density of the Stern layer (Purwidyantri et al., 2021).

	$1\times$ **PBS**	$0.1\times$ **PBS**	$0.01\times$ **PBS**
λ_D (nm)	0.76	2.41	7.61
Base pairs of B-DNA within λ_D	$\sim$2	$\sim$7	$\sim$22
Sensitivity (mV per decade)	12.67	15.95	26.69
Limit of detection (aM)	22.60	7.58	5.01

Table 10: Debye length of the three PBS dilutions, the corresponding number of base pairs of a B-form duplex (0.34 nm per base pair (Watson and Crick, 1953)), and the response of graphene transistors to the hybridisation of a 25-nucleotide target in each dilution, as reported by Purwidyantri et al. (2021).

Hybridisation itself requires salt: cations screen the electrostatic repulsion between the negatively charged backbones, so that a higher ionic strength promotes duplex formation (Campos et al., 2019). For the 30 probes selected here, the median T_m predicted by the same nearest-neighbour model falls from $58.0\,^{\circ}$C in $1\times$ PBS to $47.0\,^{\circ}$C in $0.1\times$ PBS and to $36.7\,^{\circ}$C in $0.01\times$ PBS, below the hybridisation temperature of $37\,^{\circ}$C; the last value is an estimate, since it extrapolates the salt correction to a very low ionic strength. Diluting a sample to lengthen λ_D also dilutes the analyte (Kesler et al., 2020). The buffer in which the device is read, however, need not be the one in which hybridisation takes place: Campos et al. (2019) hybridised in a buffer of high ionic strength (10 mM phosphate buffer, 150 mM NaCl and 50 mM $MgCl_2$) and measured in 10 mM phosphate buffer after a stringency rinse. The probes are therefore designed for $1\times$ PBS, and the readout buffer remains a parameter of the device protocol.

Strategies that achieve detection at physiological ionic strength act on the device rather than on the probe sequence: a biomolecule-permeable layer of polyethylene glycol on the sensing surface (Gao et al., 2015), also applied to graphene transistors (Gao et al., 2016), or the transduction of a target-induced conformational change that occurs within the screening length (Nakatsuka et al., 2018). Within the scope of probe design, $1\times$ PBS is thus taken as the physiologically meaningful reference, and the more dilute buffers as readout scenarios evaluated for sensitivity.

4.10 Debye coverage of the interface models

The models make measurable a device-relevant quantity that no sequence-level metric captures: how much of the probe, and of its charge, lies close enough to the graphene to be electrically visible (Section 3.11.3). Three geometries were measured — the rigid placement, the probe anchored by its $3'$ end,

and the anchored duplex — in the three electrolytes, whose Debye lengths differ by an order of magnitude. Figure 9 shows what each geometry represents, with the measured counts that the following sections report.

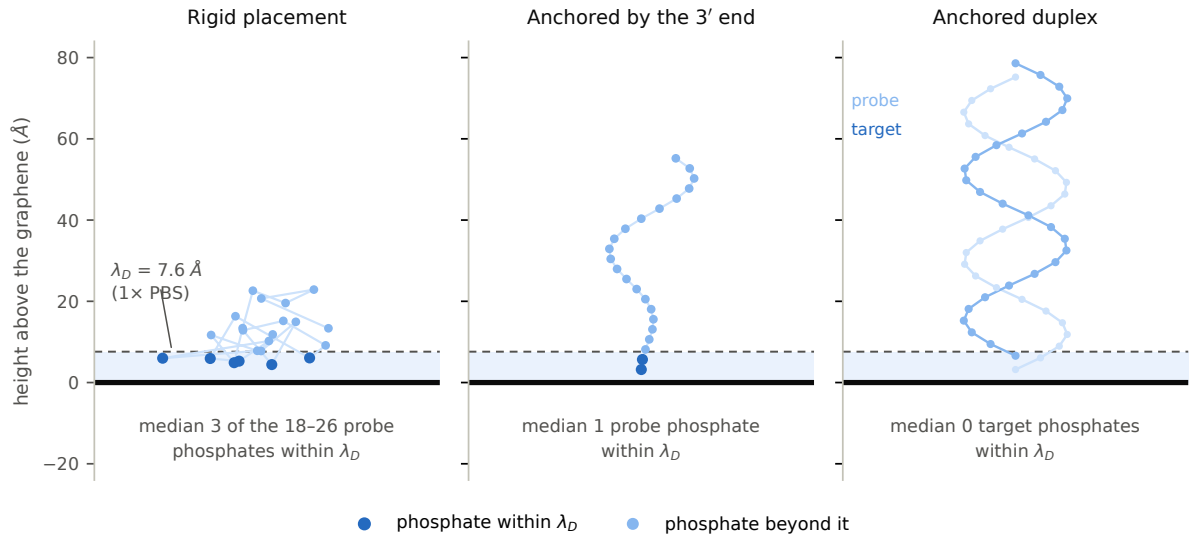


Figure 9: Geometries of the Debye measurement, drawn against the screening length at $1\times$ PBS (shaded band). Left: the rigid placement, which preserves the orientation predicted by Boltz-2 and rests the lowest atom at the contact distance. Centre: the probe tethered by its $3'$ end, as it is immobilised experimentally. Right: the probe hybridised to its target, the target running antiparallel so that its first phosphate sits one helical rise above the anchor. The conformations are schematic; the counts beneath each panel are the medians measured over the 30 models.

4.10.1 Rigid placement

In the placement that preserves the predicted orientation, the fraction of probe atoms within λ_D ranges from 2.3 to 14.2% (median 7.4%) in $1\times$ PBS, which corresponds to between 0 and 4 of the 18–26 phosphates of each probe (median 3). In $0.1\times$ PBS the fraction rises to 23.2–97.8% (median 58.9%), and in $0.01\times$ PBS it reaches 100% for every model: the tallest probe stands 62.4 Å above the sheet, well inside the 76.1 Å of λ_D at that dilution.

Table 11 breaks these values down by Boltz-2 confidence class. At $1\times$ PBS no model of any class reaches even 15% coverage, and the class medians are ordered *against* the confidence (LOW 8.5%, MODERATE 6.2%, GOOD 5.6%). At $0.1\times$ PBS the models separate, from 23.2% to 97.8%, and it is in this buffer that the criterion is most informative: 22 of the 30 models place at least half of their atoms inside the screening length. At $0.01\times$ PBS the criterion saturates, assigning 100% to every candidate and therefore

distinguishing none.

		Height	Probe atoms within λ_D			
Class	n	**(Å)**	$1\times$	$0.1\times$	$0.1\times$, $\geq 50\%$	$0.01\times$
GOOD	4	47.6	5.6%	53.9%	2/4	100.0%
MODERATE	13	40.1	6.2%	56.5%	10/13	100.0%
LOW	13	33.4	8.5%	73.4%	10/13	100.0%
All	**30**	**38.2**	**7.4%**	**58.9%**	**22/30**	**100.0%**

Table 11: Debye coverage of the 30 models by Boltz-2 confidence class. "Height" is the median height of the probe above the graphene plane; the coverage columns give the median percentage of probe atoms within λ_D in each buffer, and the sixth column the number of models of each class with at least half of the probe inside λ_D at $0.1\times$ PBS.

The correlation between coverage and 3D confidence is weak and slightly negative in both buffers in which it can be computed ($r = -0.19$ at $1\times$ PBS and $r = -0.14$ at $0.1\times$ PBS; at $0.01\times$ PBS the coverage is the same for every probe, so no correlation is defined), so a probe that is better predicted is not thereby more visible to the device. This is consistent with the independence already observed between the 3D confidence and the sequence-level quality (Section 4.3), and it is what makes interface geometry worth carrying as a criterion of its own: it orders the candidates along an axis that neither the sequence screening nor the structure prediction captures.

4.10.2 Anchored probe and hybridised duplex

The rigid placement inherits the orientation predicted for the free probe, which is arbitrary with respect to the surface. The anchored geometry removes that arbitrariness by reproducing the immobilisation used experimentally — the probe tethered by its $3'$ end (Campos et al., 2019; Purwidyantri et al., 2021) — and by averaging over the orientations such a tether allows. Of the $20\,000$ orientations sampled per probe, those retained (median 5339 per probe) place a median of 1 to 2 phosphates within λ_D in $1\times$ PBS for 28 of the 30 probes, and at most 8 in the most favourable orientation of any probe. Anchoring therefore reduces the charge inside the screening length relative to the unconstrained placement, as expected: the probe can no longer present whichever face happens to be closest to the surface.

The number of retained orientations varies widely between probes, from 98 to 6331, and it is itself informative. It measures how exposed the $3'$ terminus is in the predicted conformation: an orientation is

retained only when that terminus is the part of the molecule closest to the surface, which is rare when it is tucked into the body of the structure. Across the panel the retained count is strongly and negatively related to the anchored charge ($r = -0.89$; Spearman $\rho = -0.85$), and the relation has a geometric reading rather than being an artefact of sampling. When the $3'$ end protrudes, many orientations qualify, but in them the rest of the probe extends away from the contact point and few phosphates remain within λ_D. When it is buried, few orientations qualify, and in those few the molecule is necessarily folded around its own anchor, so more of it stays close to the surface. The two probes with fewest retained orientations, *p2331* (98) and *p4010* (542), are accordingly those with the highest anchored charge (5 and 4 phosphates), while the probes with more than 4000 retained orientations have a median of 1.

This has two consequences. The anchored metric is not a property of the sequence alone: it is governed largely by where the $3'$ terminus happens to sit in the predicted conformation, which is itself a prediction. And the estimates for probes with few retained orientations rest on a small sample, so their values are the least reliable of the panel — a caveat that applies precisely to the probes the metric favours.

The hybridised state is the one the device actually senses, since it is the arrival of the target that changes the charge at the surface. Measured on the anchored duplexes, in $1\times$ PBS the median number of target phosphates within λ_D is zero for all 30 probes, and even the most favourable orientation places only 4–7 of them inside. In $0.1\times$ PBS the median rises to 8 target phosphates, and in $0.01\times$ PBS the entire target lies inside λ_D for every probe. The charge that hybridisation brings into the sensing volume is thus set by the screening length and by the geometry of the duplex, not by the length of the probe: a longer probe captures a longer target, but the additional charge is added beyond λ_D, where it is screened. This is the quantitative counterpart, within this panel, of the dimensional mismatch discussed in Section 4.9.

4.10.3 Final selection

Because the Boltz-2 confidence and the anchored charge within λ_D are independent, the candidates of each gene were compared on both axes at once and those on the Pareto front retained (Section 3.11.3). The front contains 11 of the 30 probes (Table 12), between one and three per gene. Its composition shows why neither axis alone would suffice: it includes the probes with the highest confidence of their gene, such as *p598* (*nuc*) and *p5178* (*algD*), but also probes of low confidence that present appreciably more charge at the interface, such as *p4010* (*oprL*) and *p2331* (*lytA*). Ranking by confidence alone would have discarded the latter; ranking by coverage alone would have discarded the former.

The two probes admitted on charge alone, *p4010* and *p2331*, are those whose anchored values rest on the fewest sampled orientations, for the reason given above. Restricting the front to the 28 probes with at

least 1000 retained orientations leaves 9 probes and removes exactly those two, without changing any other entry. Both lists are defensible, and the difference between them is reported here rather than resolved: the longer list is the one the stated criterion produces, while the shorter one is the more conservative reading. Either way, every gene keeps at least one candidate, and the probes selected on confidence are unaffected. Figure 10 shows the front of each gene against the candidates it dominates.

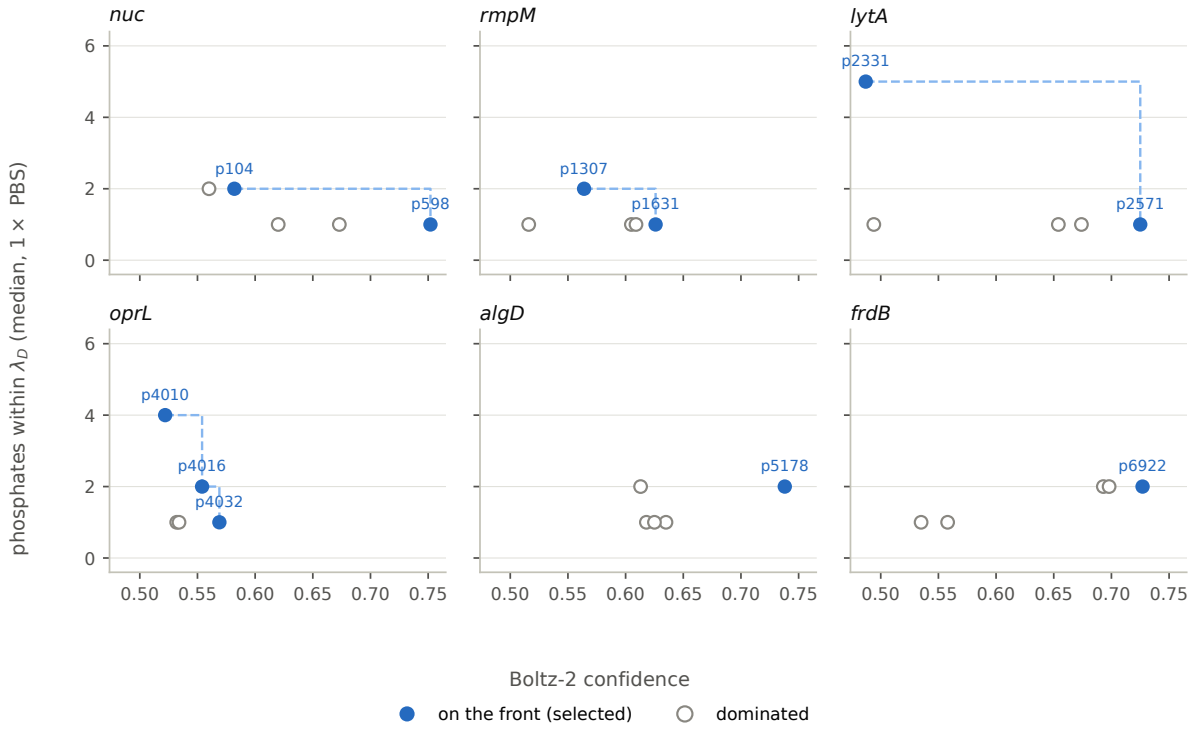


Figure 10: Boltz-2 confidence against the median number of anchored phosphates within λ_D in $1\times$ PBS, for the five candidates of each gene. Filled points lie on the Pareto front and enter the final selection; hollow points are dominated, that is, another candidate of the same gene is at least as good on both axes and better on one. The dashed line traces the front. Because the two axes are uncorrelated across the panel (Section 4.3), the fronts are not single points: *oprL* retains three candidates and *algD* and *frdB* one each.

Neither list has been screened for cross-hybridisation with organisms outside the target panel. The probes on it are conserved within their own gene, which is a criterion of sensitivity; whether they also match sequences they should not is a separate question that this work does not answer, and it is set out in Section 4.11.

Gene	Probe	Class	Confidence	Phosphates within λ_D		Orientations
				median	max.	
nuc	p598	GOOD	0.752	1	6	6293
nuc	p104	LOW	0.582	2	5	2337
rmpM	p1631	MODERATE	0.626	1	5	5413
rmpM	p1307	LOW	0.564	2	5	1779
lytA	p2571	GOOD	0.725	1	6	5675
lytA	p2331	LOW	0.487	5	7	98
oprL	p4032	LOW	0.569	1	5	5941
oprL	p4016	LOW	0.554	2	5	1623
oprL	p4010	LOW	0.522	4	5	542
algD	p5178	GOOD	0.738	2	6	4010
frdB	p6922	GOOD	0.727	2	6	3959

Table 12: Final selection: the probes on the Pareto front of Boltz-2 confidence and anchored phosphates within λ_D in $1\times$ PBS, by gene. "Orientations" is the number of sampled orientations retained for the probe.

4.11 Limitations

The following considerations bound how far the results reported above can be taken.

Specificity is not verified. The selection is driven by conservation, which is a criterion of sensitivity: it requires that a probe match every strain of its target. It never asks the complementary question, whether the probe also matches something it should not. A region may be perfectly conserved within *S. aureus* and still be shared with a related species, or with the human genome that is present in any clinical sample, and nothing in the pipeline would detect it. The candidates reported here are therefore candidates for specificity testing rather than a panel ready for use, and a BLAST-based specificity screen against non-target genomes and the host is the first item of future work.

No experimental validation. Every result is computational. No probe was synthesised, no device was functionalised and no hybridisation was measured. The experimental quantities that appear in this

dissertation are taken from published work on comparable devices (Campos et al., 2019; Purwidyantri et al., 2021) and serve to frame the computational results, not to confirm them. The structures are likewise predictions: the confidences obtained for the single strands are modest (Section 4.3), and although the duplexes are predicted with high confidence and canonical geometry, no experimental structure of these probes exists against which to check them.

The sequence sample is what NCBI contains. Conservation is measured over deposited records, with the sampling bias described in Section 4.4, and the retrieval depends on queries by gene name and organism, which will miss records annotated differently. Both affect which regions are judged conserved.

The interface models are idealised. The rigid placement keeps the orientation predicted by Boltz-2 and is not optimised against the surface. Since single-stranded DNA adsorbs flat on graphene through π–π stacking (Green and Norton, 2015), the orientations reached after docking or molecular dynamics will differ, and so will the coverage. The anchored geometry is less exposed to that objection, because it averages over orientations rather than trusting one, but it idealises the tether in turn: no linker length is added, so the $3'$ end sits as close to the sheet as it can, and the coverage is an upper bound. It also depends on where the $3'$ terminus falls in the predicted conformation, as Section 4.10 shows. The models further omit the pyrene linker itself, the surface density of probes, which in practice is high enough for neighbours to interact, and the blocking agent used to passivate unreacted linker.

The coverage is a geometric index. It follows from the Debye–Hückel picture, which overstates the practical penalty of a concentrated buffer, as the measured response of the same devices shows (Section 4.9). The values serve to compare candidates with one another, which is how they enter the selection, and not to predict whether a device will respond.

4.12 Curated dataset of prioritised candidates

Finally, the same quality ranking was used to export a larger curated dataset: the top 25 probes per gene (150 probes in total), each with its sequence-level features (T_m, GC, conservation, No-fold and folding energies). This dataset extends the set of 30 probes predicted in 3D to a broader, biophysically annotated selection of the highest-quality candidates per gene.

Its purpose is to keep the work usable beyond the eleven probes finally selected. The structural and interface analyses are expensive and were therefore applied to five candidates per gene, but the selection

narrows at every stage and a candidate can fail for reasons this pipeline does not test, specificity above all (Section 4.11). A ranked pool of 150 annotated candidates means that the next one in line for a gene is already characterised at the sequence level and can be carried forward without rerunning the retrieval, the alignment or the screening. The same table is also the feature matrix that a data-driven model of probe quality would be trained on, which is among the directions set out in Chapter 5.

Chapter 5

Conclusions and Future Work

5.1 Conclusions

This dissertation set out to design, *in silico*, high-quality **ssDNA** capture probes for **GFET** biosensors targeting a panel of clinically relevant bacterial pathogens. The main outcome is a single, reproducible computational pipeline that turns a target gene into a small set of ranked, biophysically screened probe candidates, with every design decision grounded in the literature.

The pipeline integrates, in one automated workflow, the stages that are usually performed separately or by hand: retrieval and curation of sequences from **NCBI**, automatic length clustering, multiple sequence alignment and **PPI**-based conservation scoring, nearest-neighbour thermodynamic screening, secondary-structure evaluation with a continuous No-fold score, and a final three-dimensional structure prediction with Boltz-2. Two design choices give the method its generality and rigour. First, thresholds are resolved through a layered, *organism-aware* scheme (gene $\rightarrow$ species $\rightarrow$ type $\rightarrow$ global), so that pathogens with very different genome composition are judged by fair criteria; this was shown to recover candidates — such as AT-rich probes — that a single global rule would wrongly discard. Second, candidates are prioritised by a transparent quality score combining conservation and the No-fold score, which concentrates the expensive 3D prediction on the most promising sequences.

Applied to the six-gene panel, with every thermodynamic quantity computed in $1\times$ PBS at $37\,^{\circ}\mathrm{C}$ — the buffer whose ionic composition is closest to that of physiological fluids — the pipeline reduced 7661 candidate windows to 2439 that passed screening, from which the five best candidates per gene were carried into three-dimensional prediction. Alignment-free diversity and rarefaction analyses characterised each target and justified the sampling depth in a principled, non-arbitrary way. An existing experimental probe panel was folded into the same workflow and scored with identical metrics, which placed the designed probes against sequences the method did not select, and a curated 150-probe dataset of the highest-quality candidates per gene was produced. Screening in the reference buffer changed which candi-

dates survive, and it is what gives the melting temperature, hairpin and homodimer thresholds a defensible meaning, since they then refer to the solution in which the probes are meant to work.

The selected probes were predicted with Boltz-2 both as single strands and as duplexes with their complementary targets, and assembled with a graphene sheet into atomistic models of the sensing interface, using a pristine layer built with GOPY and a placement set at the equilibrium physisorption distance reported for nucleobases on graphene. All 30 models are free of steric clashes, and three findings emerged from them.

First, the three-dimensional prediction is not a restatement of the sequence ranking: across the 30 probes its confidence is uncorrelated with the quality score used to select them ($r = -0.01$), with the conservation and with the No-fold score. Second, the fraction of each probe within the Debye length is likewise uncorrelated with that confidence, and it varies widely across the panel — from 2.3 to 14.2% at physiological ionic strength and from 23 to 98% in tenfold-diluted buffer. Interface geometry is therefore an independent axis of probe quality, and the final selection was accordingly made on the Pareto front of the two criteria, retaining 11 of the 30 probes, between one and three per gene. Modelling the probe as it is actually immobilised, tethered by its $3'$ end, also revealed a limitation of that criterion: the charge it places within the Debye length depends strongly on where the $3'$ terminus lies in the predicted conformation ($r = -0.89$ against the number of admissible orientations), so the probes the criterion favours are also those whose estimate rests on the fewest sampled orientations. Third, modelling the hybridised state showed that at physiological ionic strength the median number of target phosphates inside the Debye length is zero for every probe of the panel: the detectable charge is set by the screening length and the geometry of the duplex, not by the length of the probe. This is a limitation of the transduction rather than of these particular sequences, and it is consistent with the experimental observation that such devices still detect hybridisation in $1\times$ PBS, at roughly half the sensitivity obtained in dilute buffer.

The work also produced a result that was not among the original objectives: a measure of how much of a probe, and of its hybridised target, lies within the sensing volume of the device, together with the finding that this quantity is independent of both the sequence-level ranking and the structural confidence.

Overall, the work delivers an extensible foundation for the computational stage of **GFET** biosensor development. It produces prioritised probe candidates under conditions that match the intended application and generalises to new genes and organisms with minimal configuration, within the limits set out in Section 4.11.

5.2 Prospects for future work

Several directions follow naturally from this work:

- **Experimental validation.** The ultimate test of the designed probes is their performance on fabricated **GFET** devices; the ranked shortlist provides the immediate candidates for functionalisation and characterisation.
- **Specificity screening.** Adding an automated specificity check (for example a BLAST step against non-target genomes and the host) would complement the current conservation-driven, sensitivity-oriented selection.
- **Refinement of the interface models.** The assembled models are starting geometries. Molecular docking and force-field refinement would relax the probe against the surface and replace the orientation sampling used here by energetically weighted poses, from which the Debye coverage could be recomputed.
- **Data-driven modelling.** The curated feature dataset opens the way to a machine-learning model that predicts probe quality directly, potentially accelerating or refining the ranking.
- **Broader panels.** As the framework already supports viral, fungal, protozoan and host profiles, extending the panel to further genes and non-bacterial pathogens is largely a matter of configuration.

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